

Steganography and Image File Forensics

Module Objectives

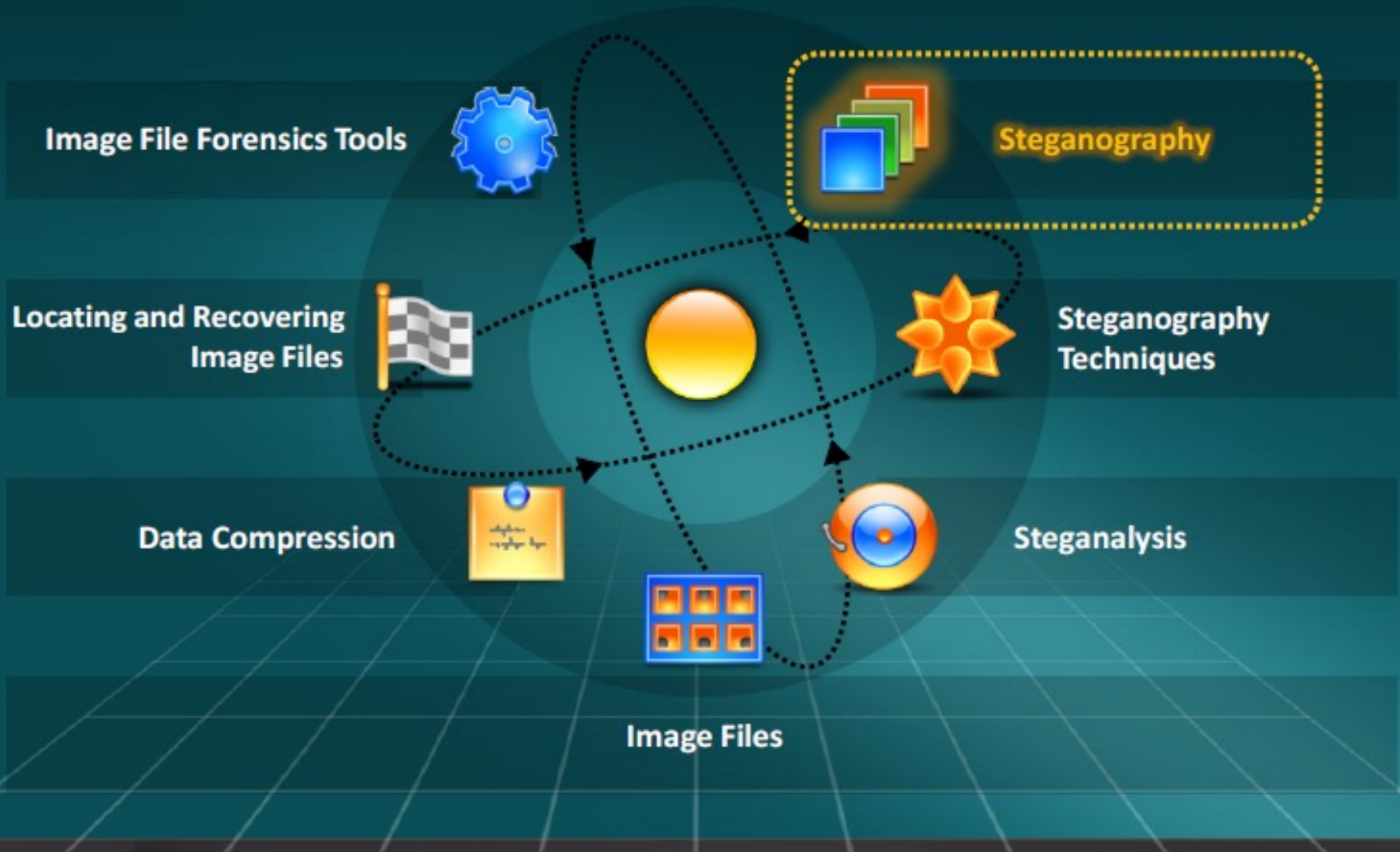
- What is Steganography?
- Application of Steganography
- Steganography Techniques
- Classification of Steganography
- Types of Steganography
- Steganalysis
- How to Detect Steganography
- Steganography Detection Tools



- Image Files
- Understanding Image File Formats
- Data Compression
- Forensic Image Processing Using MATLAB
- Locating and Recovering Image Files
- Identifying Unknown File Formats
- Picture Viewer Tools
- Image File Forensic Tools



Module Flow



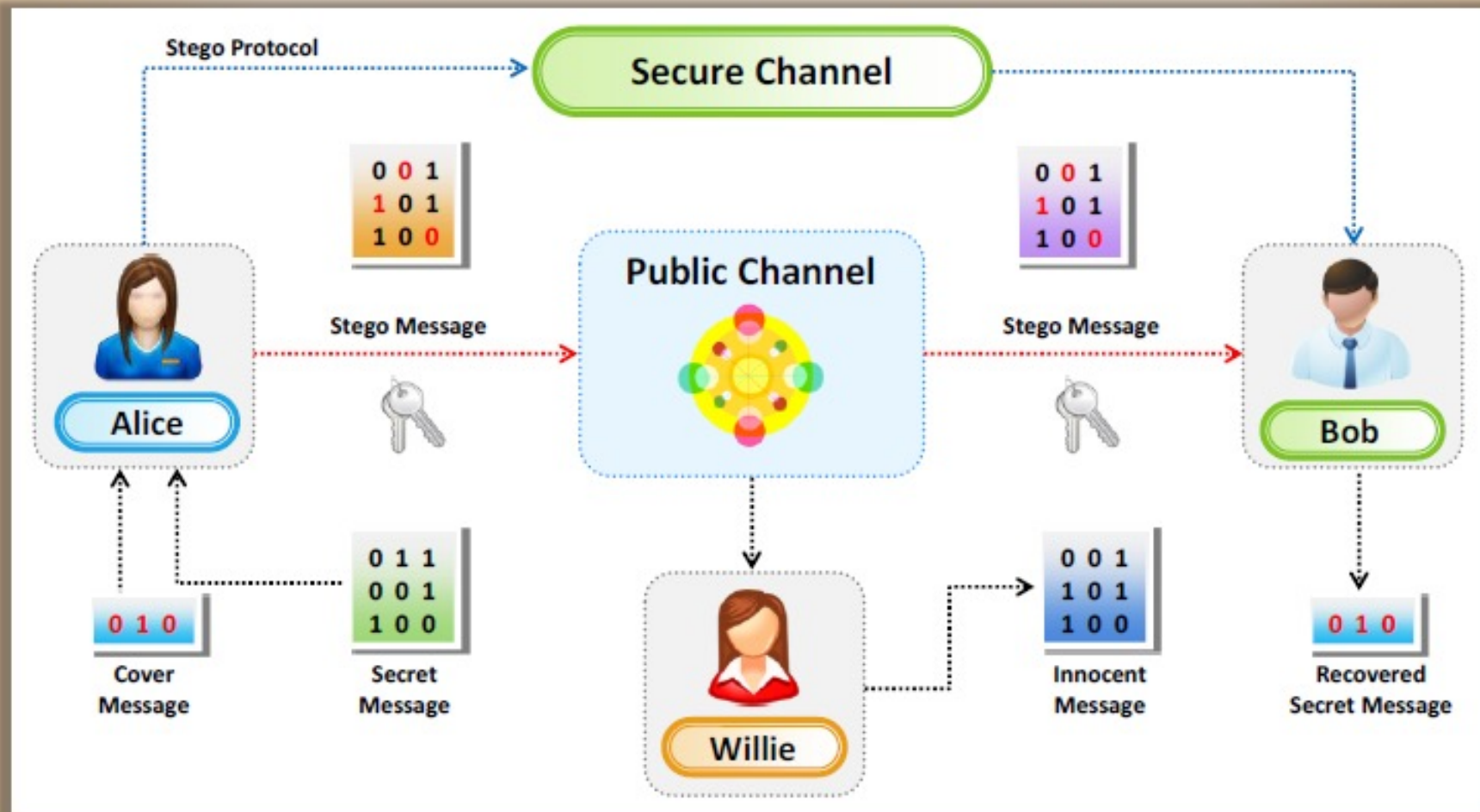
What is Steganography?

- Steganography is a technique of **hiding a secret message** within an ordinary message and **extracting it at the destination** to maintain confidentiality of data
- **Utilizing a graphic image as a cover** is the most popular method of concealing data in files



How Steganography Works

Stegosystem describes the process that is used in performing steganography



Legal Use of **Steganography**

Law enforcement agencies use steganography to:



Steganographically watermark intermediation materials after **authorization** and under public prosecutor control with predefined marks

Trace trade materials



Build an international **data bank** to collect data on the trading controlled by investigative bodies

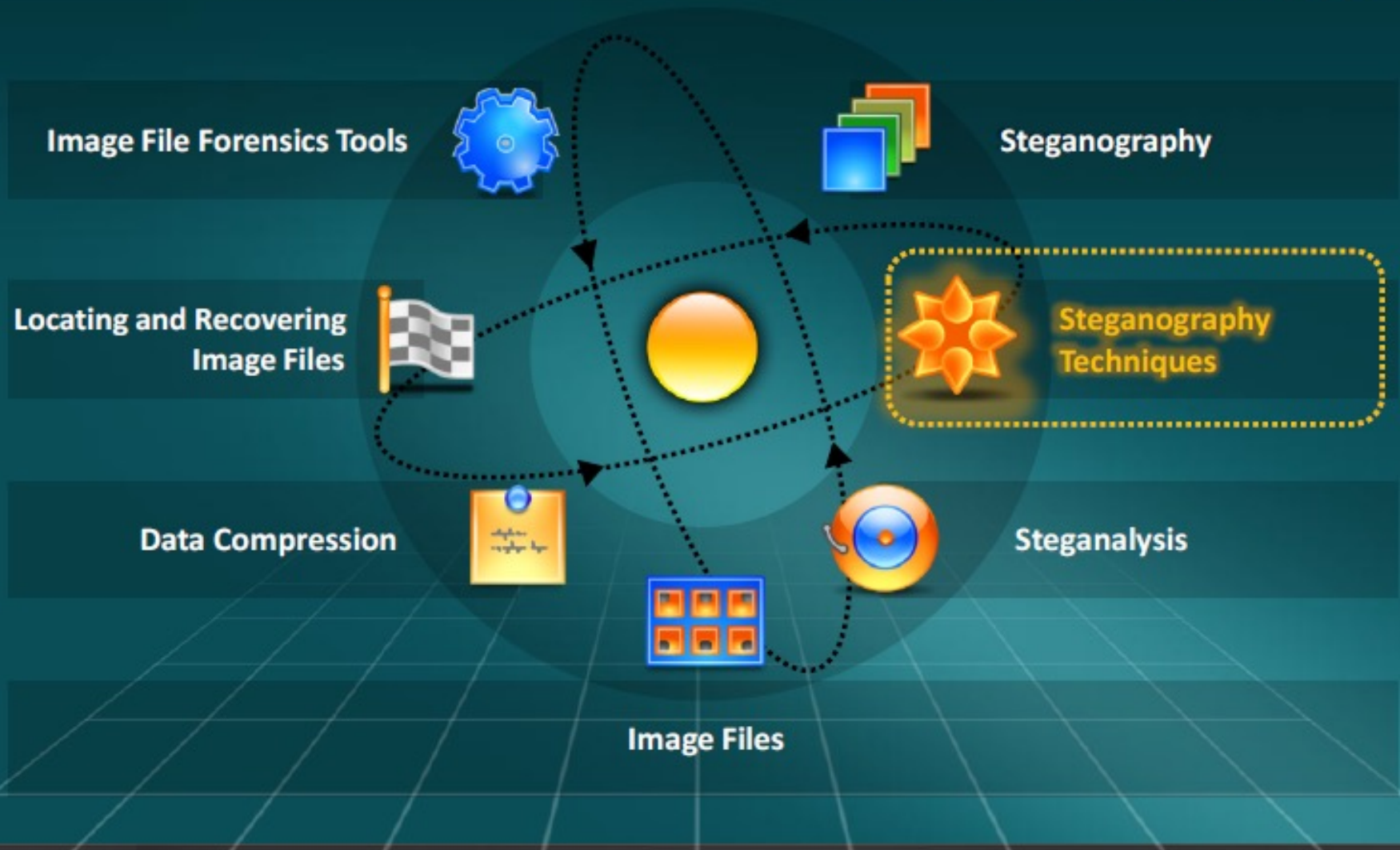
Provide **network nodes** where trade material is monitored



Unethical Use of Steganography



Module Flow



Steganography Techniques



Substitution Techniques

Substitute redundant part of the cover object with a secret message

Transform Domain Techniques

Embed secret message in a transform space of the signal (e.g. in the frequency domain)



Cover Generation Techniques

Encode information that ensures creation of cover for secret communication

Spread Spectrum Techniques

Adopt ideas from spread spectrum communication to embed secret messages



Distortion Techniques

Store information by signal distortion and in the extraction step measure the deviation from the original cover

Statistical Techniques

Embed messages by altering statistical properties of the cover objects and use hypothesis methods for extraction



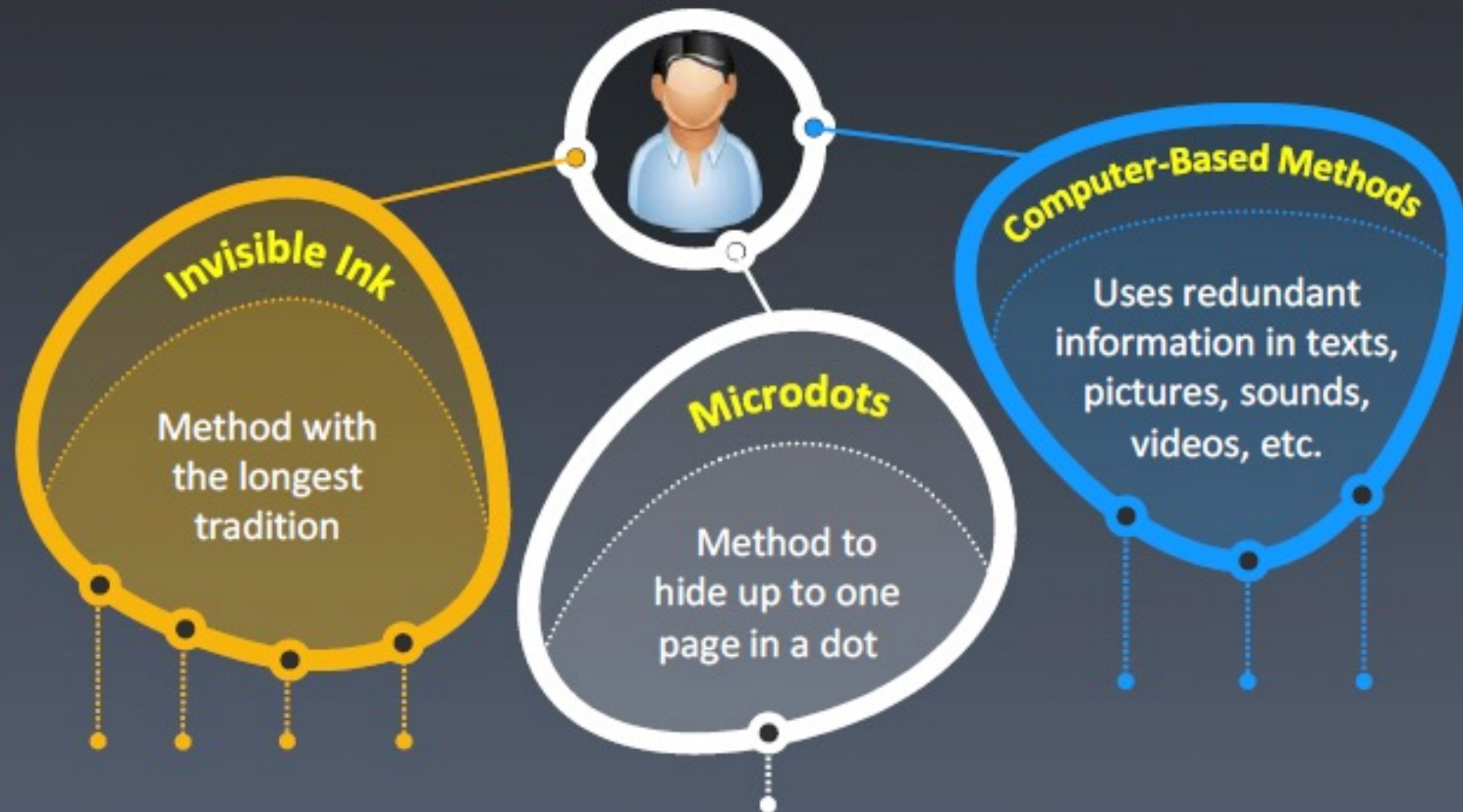
Application of **Steganography**

Steganography is applicable to the following areas:



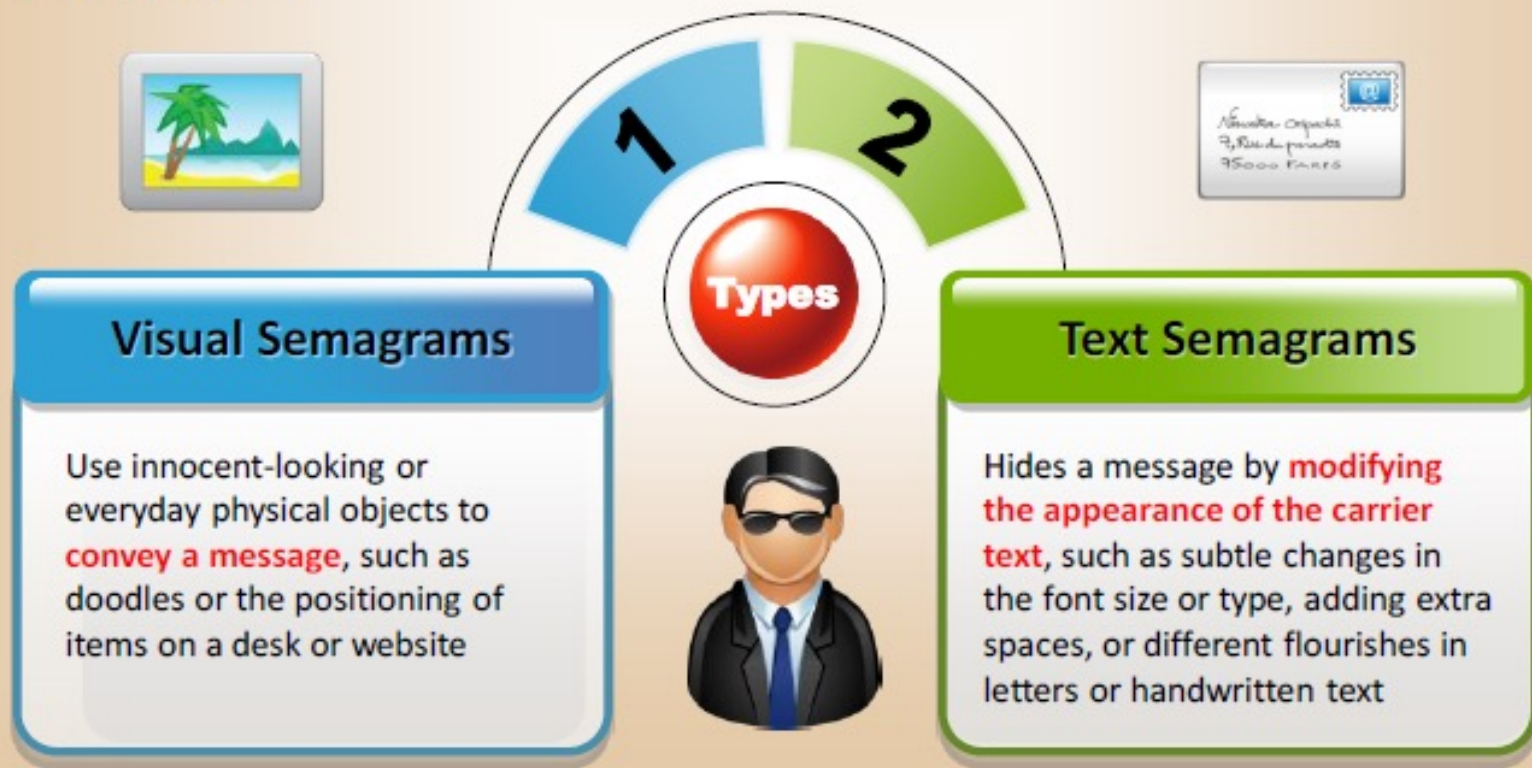
Technical Steganography

- Technical steganography uses physical or chemical means to **hide the existence of a message**
- Technical steganography uses tools, devices, or methods to **conceal messages**
- Some methods of **technical steganography** include:



Linguistic Steganography (Cont'd)

- Linguistic steganography utilizes written natural language to **hide the message in the carrier** in some non-obvious ways
- It is further categorized into **Semagrams and Open codes**
- Semagrams utilize **visual symbols or signs** to hide secret messages. They are classified into Visual and Text Semagrams



Linguistic Steganography

- Open code **hides the secret message** in a specifically designed pattern on the document that is unclear to the average reader

- Open code steganography is divided into:

1. Jargon Code

- It is a language that a **group of people can understand** but is meaningless to others

2. Covered Ciphers

- The message is **hidden openly in the carrier medium** so that anyone who knows the secret of how it was concealed can recover it



Covered Cipher is categorized into null ciphers and grille ciphers

1. Null Ciphers

- A null cipher is an ancient form of **encryption where the plaintext is mixed** with a large amount of non-cipher material

- It can also be used to **hide ciphertext**

2. Grille Ciphers

- In this technique, a **grille is created** by cutting holes in a piece of paper
- When the receiver **places the grille over the text**, the intended message can be retrieved

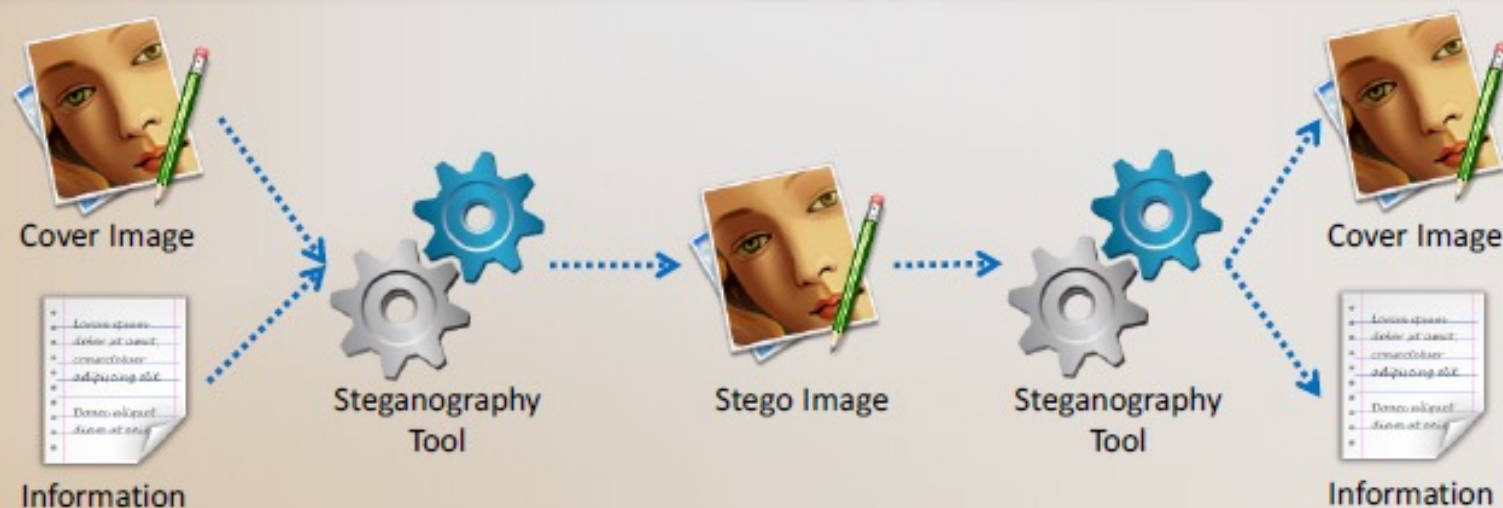


Types of Steganography



Image Steganography

1. In image steganography, the **information is hidden in image** files of different formats such as .PNG, .JPG, .BMP, etc.
2. Image steganography tools **replace redundant bits of image** data with the message in such a way that the effect cannot be detected by human eyes
3. Image file steganography techniques:
 - Least Significant Bit Insertion
 - Masking and Filtering
 - Algorithms and Transformation



Least Significant Bit Insertion

LSB

1. The **right most bit** of pixel is called the Least Significant Bit (LSB)
2. Using this method, the binary data of the **hidden message is broken** and then **inserted** into the LSB of each pixel in the image file in a deterministic sequence
3. Modifying the LSB does not result in a noticeable difference because the net change is minimal and can be indiscernible to the human eye



Example: Given a string of bytes

- (00100111 11101001 11001000) (00100111 11001000 11101001) (11001000 00100111 11101001)
- The letter "H" is represented by binary digits 01001000. To hide this "H" above stream can be changed as:
- (0010011**0** 1110100**1** 1100100**0**) (0010011**0** 1100100**1** 1110100**0**) (1100100**0** 0010011**0** 11101001)
- To retrieve the "H" combine all LSB bits **01001000**

Masking and Filtering

1

Masking and filtering techniques are mostly used on **24 bit** and **grayscale images**



2

The masking technique **hides data** using a method similar to watermarks on actual paper, and it can be done by modifying the luminance of parts of the image



3

Masking techniques hide information in such a way that the hidden message is inside the **visible part of the image**

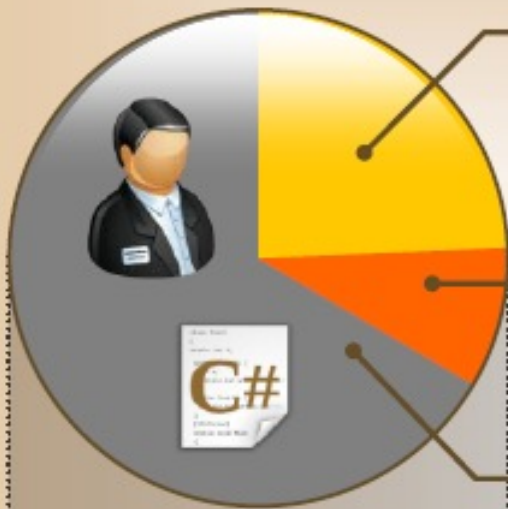


4

The information is not hidden at the **"noise"** level of the image



Algorithms and Transformation



Another steganography technique is to hide data in **mathematical functions** that are in compression algorithms

The data is embedded in the cover image by **changing the coefficients of a transform** of an image

JPEG images use the **Discrete Cosine Transform (DCT)** technique to achieve image compression

Types of transformation techniques

- Fast fourier transformation
- Discrete cosine transformation
- Wavelet transformation

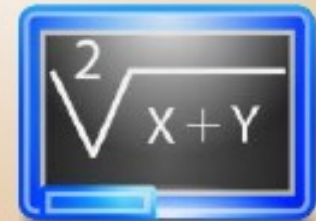


Image Steganography: Hermetic Stego

Hermetic Stego

Select operation: ☒ **Encrypt the data file and hide it in the input image(s)**
☐ Extract the data file from the input image(s) and decrypt it

☐ Select first input image ☐ Delete unsuitable input images (after confirmation)

File with data to be hidden: C:\temp\input\finances.xls Clear

Input images folder: C:\temp\input\ Clear List

Stego images folder: C:\temp\stego\ Clear List

Select first input image file

View key

Hide the data

Save configuration

Load configuration

Operation: Hide data
Data file: C:\temp\input\finances.xls
Data file size: 1,332,224 bytes
Input images folder: C:\temp\input\
Stego images folder: C:\temp\stego\
The data was successfully hidden in the following 5 images:
rock100.bmp (3,606,254 bytes)
sf_cover.bmp (2,202,678 bytes)

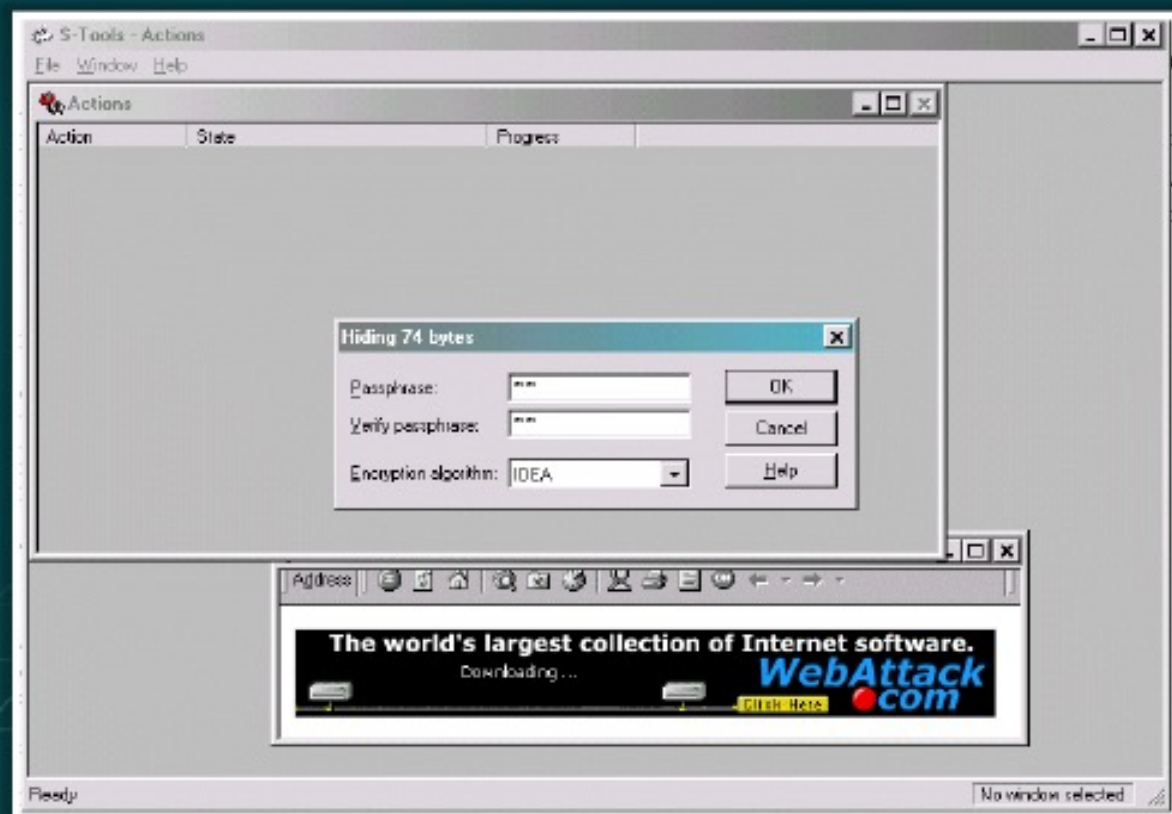
Clear Copy to clipboard Help Quit

Copyright 2003-2008 **Hermetic Systems** www.hermetic.ch Online user manual

<http://www.hermetic.ch>

Steganography Tool: S-Tools

- S-Tools can **hide multiple applications** in a single object
- It hides files in BMP, GIF, and WAV files



<http://www.spychecker.com>

Image Steganography Tools



ImageHide

<http://www.dancemammal.com>



Contraband

<http://jthz.com>



QuickStego

<http://www.quickcrypto.com>



Camera/Shy

<http://sourceforge.net>



gifshuffle

<http://www.darkside.com.au>



JPHIDE and JPSEEK

<http://linux01.gwdg.de>



OutGuess

<http://www.outguess.org>



StegaNote

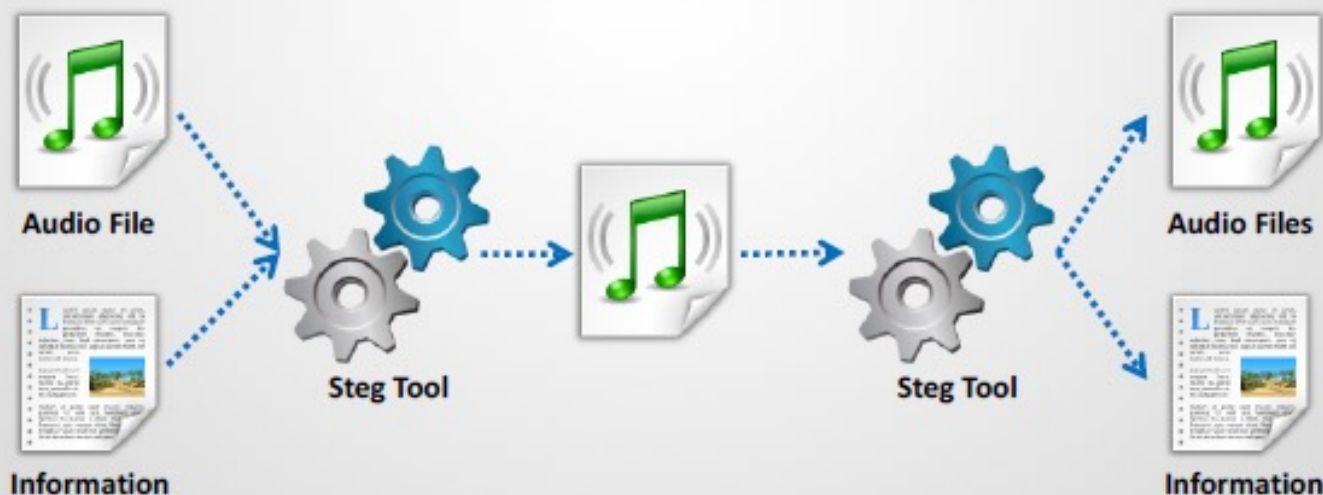
<http://www.planetsourcecode.com>





Audio Steganography

- Audio steganography refers to **hiding secret information in audio files** such as .MP3, .RM, .WAV, etc.
- Information can be hidden in an audio file by using **LSB or by using frequencies** that are inaudible to the human ear (>20,000 Hz)



Audio Steganography Methods (Cont'd)

Echo Data Hiding

- ❌ In echo data hiding technique, the **secret message** is embedded into a cover audio signal as an echo
- ❌ Parameters of the echo of the cover signal, namely **amplitude**, **decay rate** and **offset** from the original signal, are varied to represent an encoded secret binary message
- ❌ An echo cannot be easily resolved because the parameters are set below **levels audible to human**



Spread Spectrum Method

- ✅ It encodes data as a **binary sequence** that sounds like noise but can be recognized by a receiver with the correct key
- ✅ Two approaches are used in this technique, namely **direct sequence spread spectrum (DSSS)** and **frequency hopping spread spectrum (FHSS)**
- ✅ In DSSS, the secret message is spread out by **chip rate (constant)** and then modulated with a pseudo-random signal that is then interleaved with the cover signal
- ✅ In FHSS, the audio file's **frequency spectrum is altered** so that it hops rapidly between frequencies
- ✅ Spread spectrum method plays a major role in **secure communications** – commercial and military



Audio Steganography Methods

LSB Coding

- The low bit encoding technique is similar to the **least bit insertion method** done through image files
- It replaces the LSB of information in each sampling point with a **coded binary string**



Tone Insertion

- It depends on the inaudibility of **low power tones** in the presence of significantly higher spectral components
- An indirect exploitation of the psychoacoustic masking phenomenon in the spectral domain

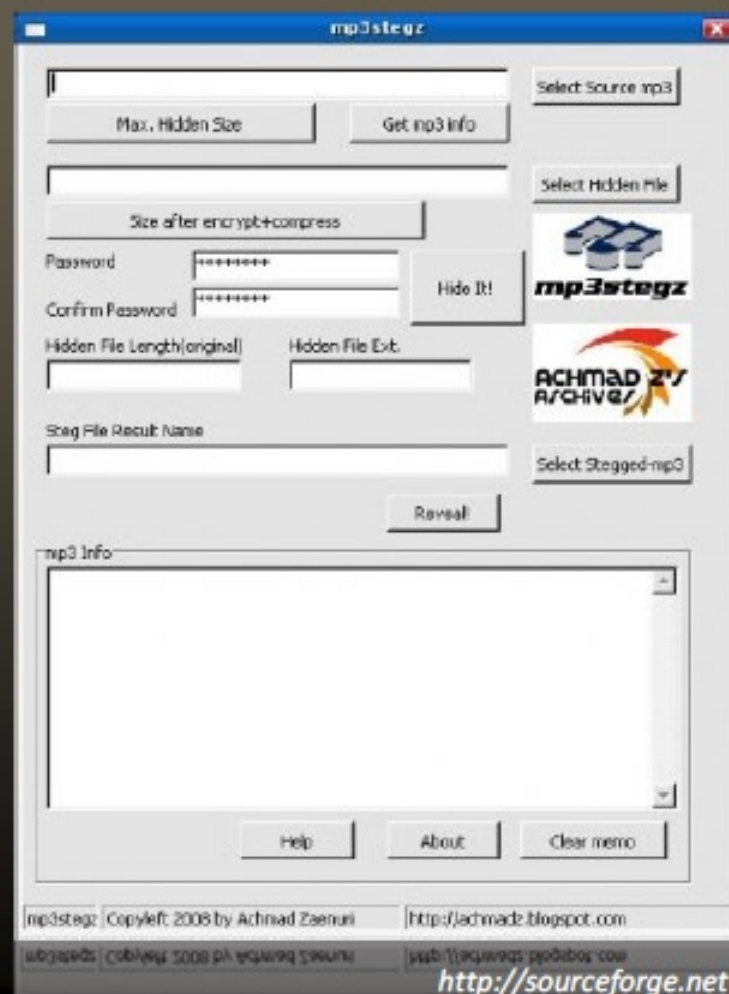


Phase Decoding

- Phase coding is described as the phase in which an **initial audio segment** is substituted by a reference phase that represents the data
- It encodes the **secret message bits** as phase shifts in the phase spectrum of a digital signal, achieving a soft encoding in terms of signal-to-noise ratio

Audio Steganography: Mp3stegz

- Mp3stegz is an application that applies a **steganographic (steganography) algorithm** in mp3 files
- It maintains the original **mp3 file's size** and **sound quality**
- The hidden message is compressed (zlib) and encrypted (Rijndael)



Audio Steganography Tools



MAXA Security Tools

<http://www.maxa-tools.com>



MP3Stego

<http://www.petitcolas.net>



Stealth Files

<http://www.froebis.com>



Steghide

<http://steghide.sourceforge.net>



Audiostegano

<http://www.mathworks.com>



Hide4PGP

<http://www.heinz-repp.onlinehome.de>



BitCrypt

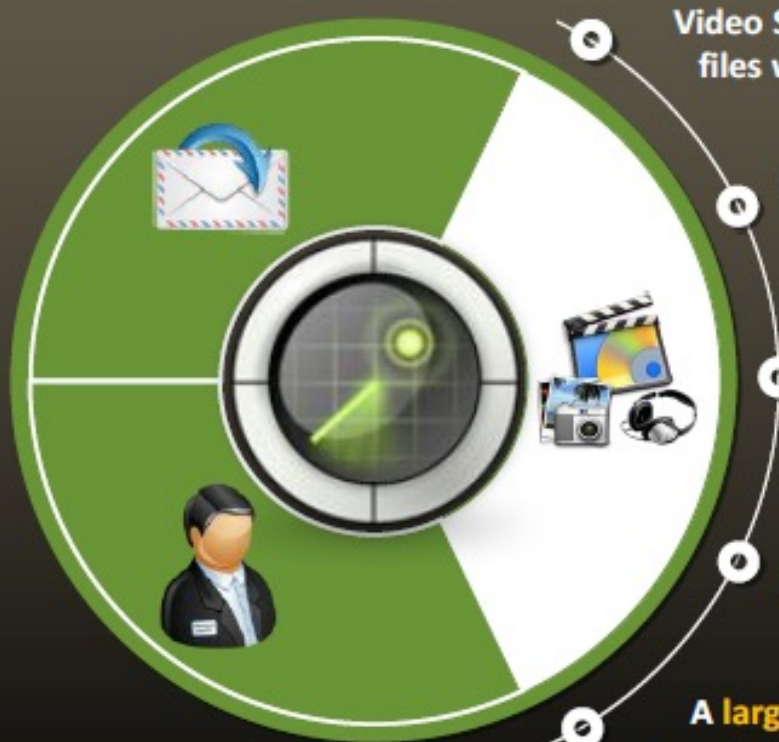
<http://bitcrypt.moshe-szweizer.com>



CHAOS Universal

<http://safechaos.com>

Video Steganography



Video Steganography refers to **hiding secret information** or any kind of files with any extension into a carrier video file

In video steganography, the information is hidden in **video files** of different formats such as .AVI, .MPG4, .WMV, etc.

Discrete Cosine Transform (DCT) manipulation is used to add secret data at the time of the transformation process of video

The techniques used in audio and image files are used in video files, as video consists of audio and images

A large number of secret messages can be hidden in video files since they are a moving stream of images and sound

Video Steganography: **MSU StegoVideo**

MSU StegoVideo allows you to hide any file in a **video sequence**

Features:

- Small video distortions after hiding info
- It is possible to extract info after video compression
- Information is protected with passcode



Video Steganography Tools



Masker

<http://www.softpuls.com>



Our Secret

<http://www.securekit.net>



Max File Encryption

<http://www.softeza.com>



BDV DataHider

<http://www.bdvnotepad.com>



Xiao Steganography

<http://xiao-steganography.en.softonic.com>



CHAOS Universal

<http://safechaos.com>



RT Steganography

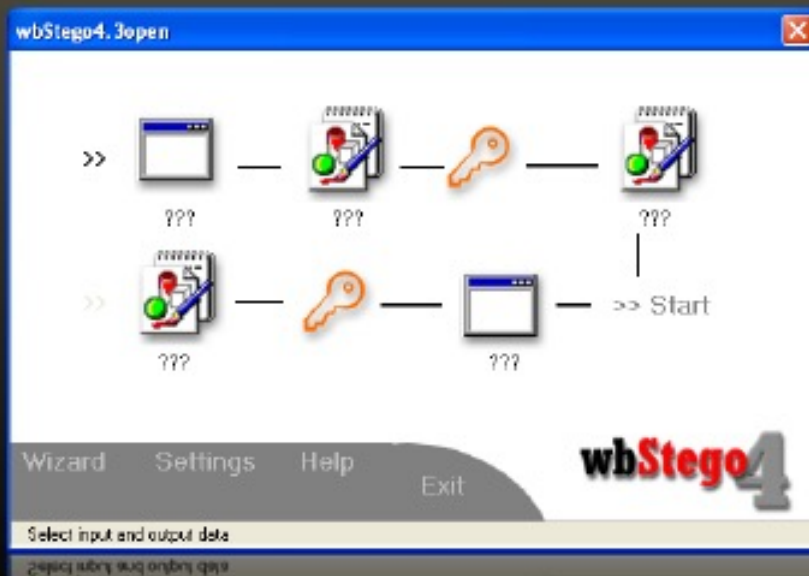
<http://sourceforge.net>



OmniHide PRO

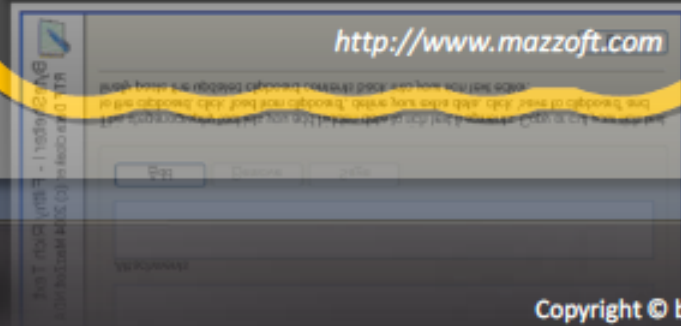
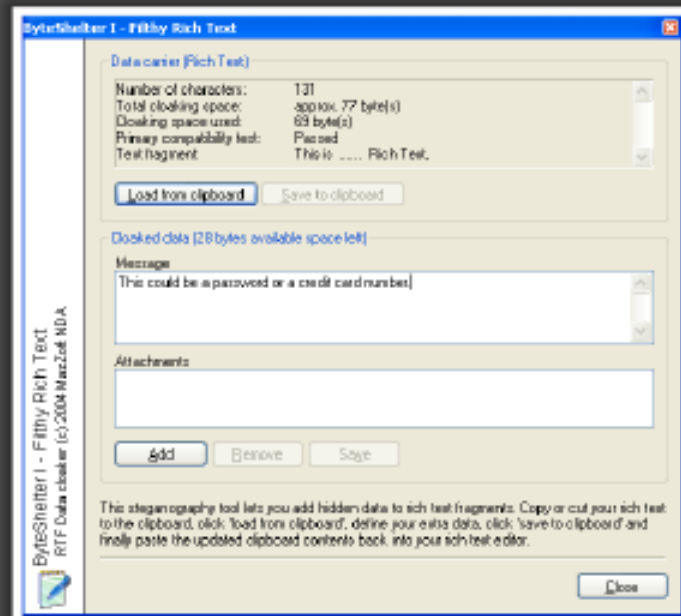
<http://omnihide.com>

Document Steganography: **wbStego**



Byte Shelter I

- Byte Shelter I **encrypts data** and **hides** it in .doc files or email messages
- Byte Shelter can be used to hide files and/or text in **rich text fragments**



Document Steganography Tools



Merge Streams

<http://www.ntkernel.com>



FoxHole

<http://foxhole.sourceforge.net>



Office XML

<http://www.irongeek.com>



Xidie Security Suite

<http://www.stegano.ro>



CryptArkan

<http://www.kuskov.com>



StegParty

<http://www.fasterlight.com>



Data Stash

<http://www.skyjuicesoftware.com>

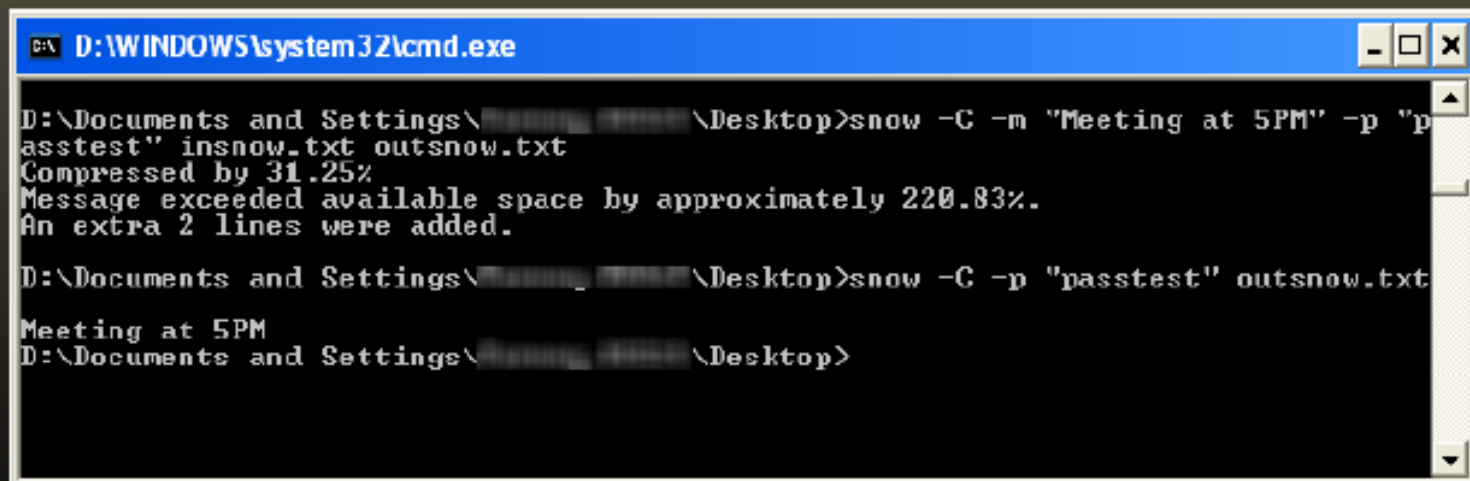


Hydan

<http://www.crazyboy.com>

Whitespace Steganography Tool: **SNOW**

1. The program SNOW is used to conceal messages in **ASCII text** by appending white space to the end of lines
2. Because spaces and tabs are generally not visible in **text viewers**, the message is effectively hidden from casual observers
3. If the **built-in encryption** is used, the message cannot be read even if it is detected



```
D:\WINDOWS\system32\cmd.exe

D:\Documents and Settings\... \Desktop>snow -G -m "Meeting at 5PM" -p "passtest" insnow.txt outsnow.txt
Compressed by 31.25%
Message exceeded available space by approximately 220.83%.
An extra 2 lines were added.

D:\Documents and Settings\... \Desktop>snow -G -p "passtest" outsnow.txt
Meeting at 5PM
D:\Documents and Settings\... \Desktop>
```

<http://www.darkside.com.au>

Folder Steganography: Invisible Secrets 4

Folder steganography refers to hiding secret information in **folders**

INVISIBLE secrets 4


Hide Files
Unhide Files
Read about the "carrier" concept...


Encrypt Files
Decrypt Files
Open Cryptboard


Shred Files
Destroy Internet Traces


IP-to-IP
Secure Password Transfer



<http://www.invisiblesecrets.com>

Invisible Secrets 4



Create Self-Decrypting Package

Files

Popup Message

Select the files you want to include in the self-decrypting package:
(You can add files by dropping them on this list). Click NEXT to continue...

Name	Type	Full Path
ewards.html	HTML Document	C:\alina\WEBSITE\
ewards.jpg	ACDSee JPEG I...	C:\alina\WEBSITE\
main.dwt.isc	Crypted File	C:\alina\WEBSITE\Templates\
main.dwt	DWT File	C:\alina\WEBSITE\Templates\
main.css	Cascading Style...	C:\alina\WEBSITE\
main.css.isc	Crypted File	C:\alina\WEBSITE\
support.jpg	ACDSee JPEG I...	C:\alina\WEBSITE\

Add Files

Add Folders

Remove

Cryptboard

Back

Next

Help

Close

CHFI
Certified
Hacking Forensic
Investigator

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Folder Steganography Tools



StegoStick

<http://stegostick.sourceforge.net>



PSM Encryptor

<http://www.powersoftmakers.com>



QuickCrypto

<http://www.quickcrypto.com>



XPTools

<http://www.xptools.net>



Max Folder Secure

<http://www.maxfoldersecure.com>



Universal Shield

<http://www.everstrike.com>



WinMend Folder Hidden

<http://www.winmend.com>



Hide My Files

<http://www.secretfilessoftware.com>

Spam/Email Steganography: Spam Mimic

Spam steganography refers to hiding information in **spam messages**



Encode

Enter your short secret message:

Alternate encodings:

- Encode as spam with a password
- Encode as fake PGP
- Encode as fake Russian
- **NEW** Encode as space

[home](#) | [encode](#) | [decode](#) | [explanation](#) | [credits](#) | [faq & feedback](#)

Encoded

Your message **Hi, I am John** gets encoded into spam as:

Dear Friend , Especially for you - this breath-taking news ! If you no longer wish to receive our publications simply reply with a Subject: of "REMOVE" and you will immediately be removed from our database . This mail is being sent in compliance with Senate bill 1622 ; Title 4 , Section 302 . This is not a get rich scheme ! Why work for somebody else when you can become rich inside 68 days . Have you ever noticed most everyone has a cellphone and how many people you know are on the Internet . Well, now is your chance to capitalize on this . We will help you decrease perceived waiting time by 170% plus increase customer response by 170% ! You can begin at absolutely no cost to you . But don't believe us , Prof Jones of Washington tried us and says "My only problem now is where to park all my cars" ! This offer is 100% legal ! For the sake of your family order now ! Sign up a friend and you'll get a discount of 50% ! Thanks !

<http://www.spammimic.com>

Steganographic File System

Steganographic file system is a method to store the files in a way that it **encrypts** and **hides the data** without the knowledge of others



It hides the user's data in other seemingly **random files**



It allows the user to **give names** and **passwords** for some files while keeping other files secret



Issues in Information Hiding

Levels of Visibility



- If the embedding process **distorts** the cover to the point that it is visually **unnoticeable**, meaning if the image is visibly distorted, then the carrier is **insufficient** for the payload. Likewise, if the image is not distorted, then the carrier is **adequate**

Robustness vs. Payload



- Redundancy is needed for a **robust** method of **embedding** the message, but it subsequently **reduces** the payload
- Robustness and payload are inversely related. Therefore, the smaller the payload, the more robust it will be

File Format Dependence



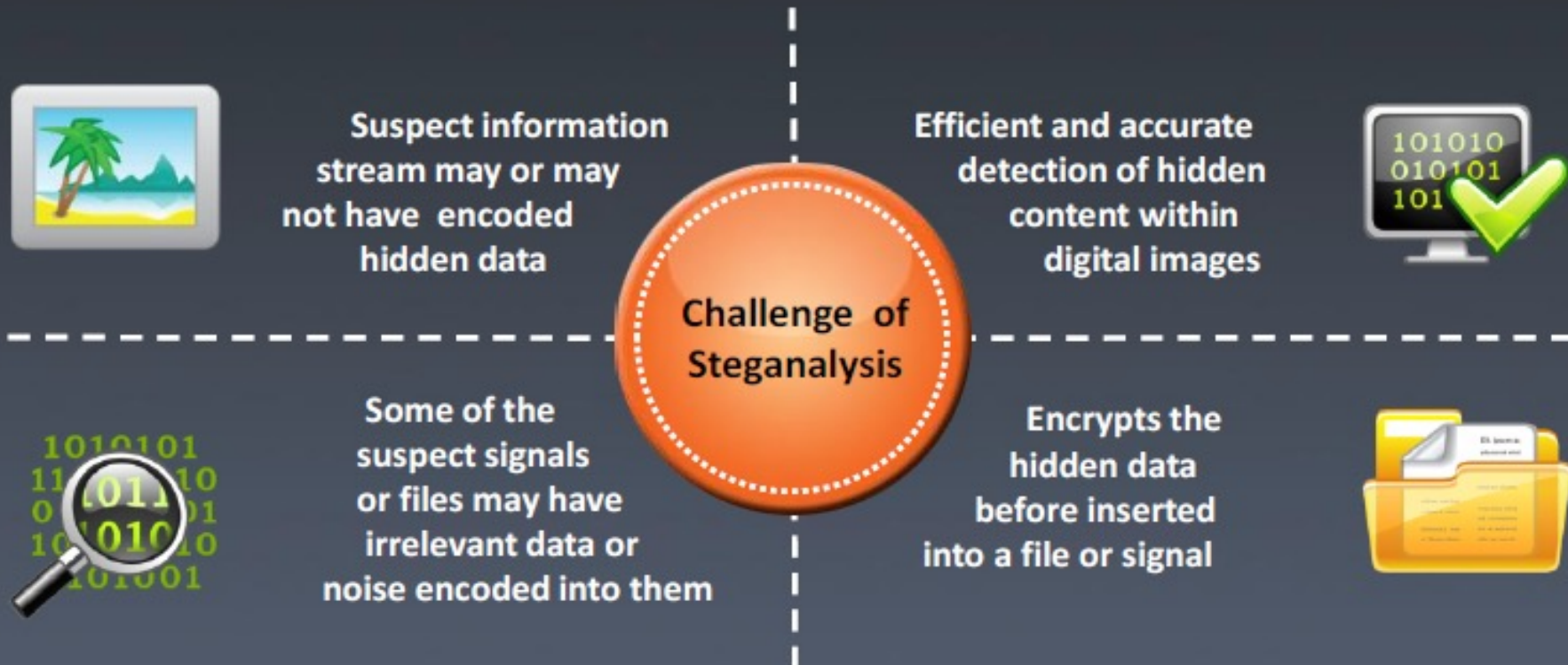
- Some image and sound files are **lossy** or **lossless**
- The conversion of lossless information to **compressed** lossy information **destroys** the **hidden information** in the cover

Module Flow



Steganalysis

Steganalysis is the art of **discovering and rendering covert messages** using steganography



How to Detect **Steganography** (Cont'd)

Software Clues on the Computer

- Steganographic investigators need to be familiar with the names of the common **steganographic software** and related terminology, and even websites about steganography
- Investigators look** for file names, website references in browser cookie/history files, registry key entries, email messages, chat/instant messaging logs, comments made by the suspect or receipts that refer to steganography
- These will provide **hard clues** for the investigator to look deeper



Investigators look for file names, website references in browser cookie/history files, registry key entries, email messages, chat/instant messaging logs, comments made by the suspect or receipts that refer to steganography

Other Program Files

- Non-steganographic software** might offer clues that the suspect hides files inside other files
- Users with binary (hex) editors, disk wiping software, or specialized chat software might demonstrate an inclination to alter files and **keep information secret**



How to Detect **Steganography**

Multimedia Files

- Look for the **presence** of a large volume of the suitable **carrier files**
- A computer system with an especially large number of files that could be **steganographic carriers** are potentially suspected
- This is particularly true if there are a significant number of seemingly **duplicate "carrier" files**



Type of Crime

- The type of crime being investigated may also make an investigator think more about **steganography** than other types of crime
- Child pornographers, for example, might use steganography to **hide their wares** when posting pictures on a website or sending them through e-mail
- Crimes that involve **business-type records** are also good steganography candidates because the **perpetrator** can hide the files but still get access to them; consider accounting fraud, identity theft (lists of stolen credit cards), drugs, gambling, hacking, smuggling, terrorism, and more



Detecting Text, Image, Audio, and Video Steganography (Cont'd)

Text File

- For the text files, the alterations are made to the **character** positions for **hiding** the data
- The alterations are detected by looking for **text patterns** or disturbances, language used, and an unusual amount of **blank spaces**



Image File

- The hidden data in an image can be detected by **determining** changes in size, file format, the last modified **timestamp**, and the **color palette** pointing to the existence of the hidden data
- Statistical analysis method is used for **image scanning**



Detecting Text, Image, **Audio**, and **Video** Steganography



Investigator

Audio File



- Statistical analysis method can also be used for audio files since the **LSB modifications** are also used on audio
- The **inaudible frequencies** can be scanned for information
- The **odd distortions and patterns** show the existence of the secret data

Video File



- Detection of the secret data in video files includes a **combination of methods** used in image and audio files
- Special code **signs** and **gestures** can also be used for detecting **secret data**

Steganalysis **Methods/Attacks** on Steganography



Only the steganography medium is available for analysis	Stego-only
Original and stego-object are available and the steganography algorithm is known	Known-stego
The hidden message and the corresponding stego-image are known	Known-message
During the communication process, active attackers can change the cover	Disabling or Active

Reformat	The format of the file is changed. This works because different file formats store data in different ways
Known-cover	The stego-object is compared with the original cover object to detect hidden information
Chosen-message	The goal is to determine patterns in the stego-object that may point to the use of the specific steganography tools or algorithms
Chosen-stego	The stego-object and steganography algorithm are identified

Disabling or Active Attacks



Blur:

It softens the **transitions** and averages the adjacent pixels with significant color change

Noise:



- Random noise **injects** random colored pixels to an image
- Uniform noise **inserts** slightly similar pixels and colors of the original pixel



Noise Reduction:

It reduces the noise in the image by **adjusting** the colors and **averaging** the pixel values



Sharpen:

It increases the **contrast** between the adjacent pixels

Rotate:

It moves the image around a **central point**



Resample

It is an **interpolation process** where the **raggedness** while expanding an image is reduced

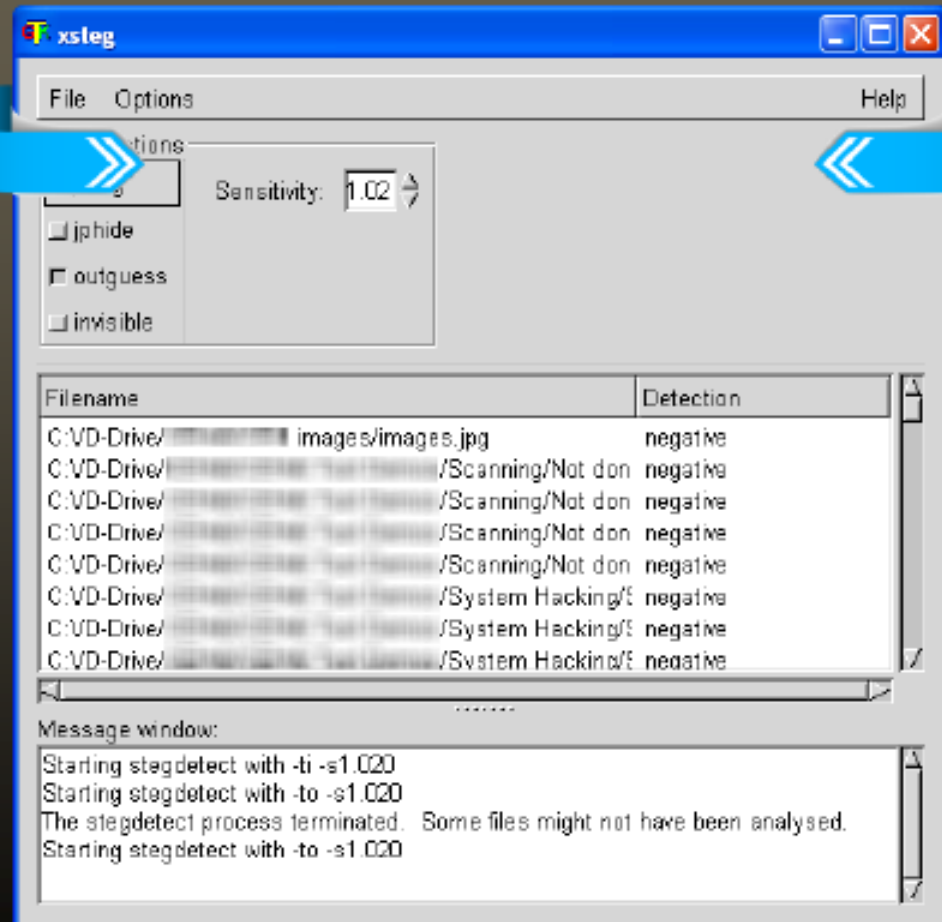


Soften:

Soften is a **uniform blur** to an image that smooths the edges and decreases the **contrasts**

Steganography Detection Tool: Stegdetect

- It is an automated tool for detecting **steganographic content** in images
- It is capable of detecting several different steganographic methods to embed **hidden information** in JPEG images



<http://www.outguess.org>

Steganography **Detection** Tools



Xstegsecret

<http://stegsecret.sourceforge.net>



StegSpy

<http://www.spy-hunter.com>



Stego Watch

<http://www.wetstonetech.com>



**Gargoyle Investigator™
Forensic Pro**

<http://www.wetstonetech.com>



StegAlyzerAS

<http://www.sarc-wv.com>



StegAlyzerSS

<http://www.sarc-wv.com>



StegAlyzerRTS

<http://www.sarc-wv.com>



StegMark

<http://www.datamark-tech.com>

Module Flow

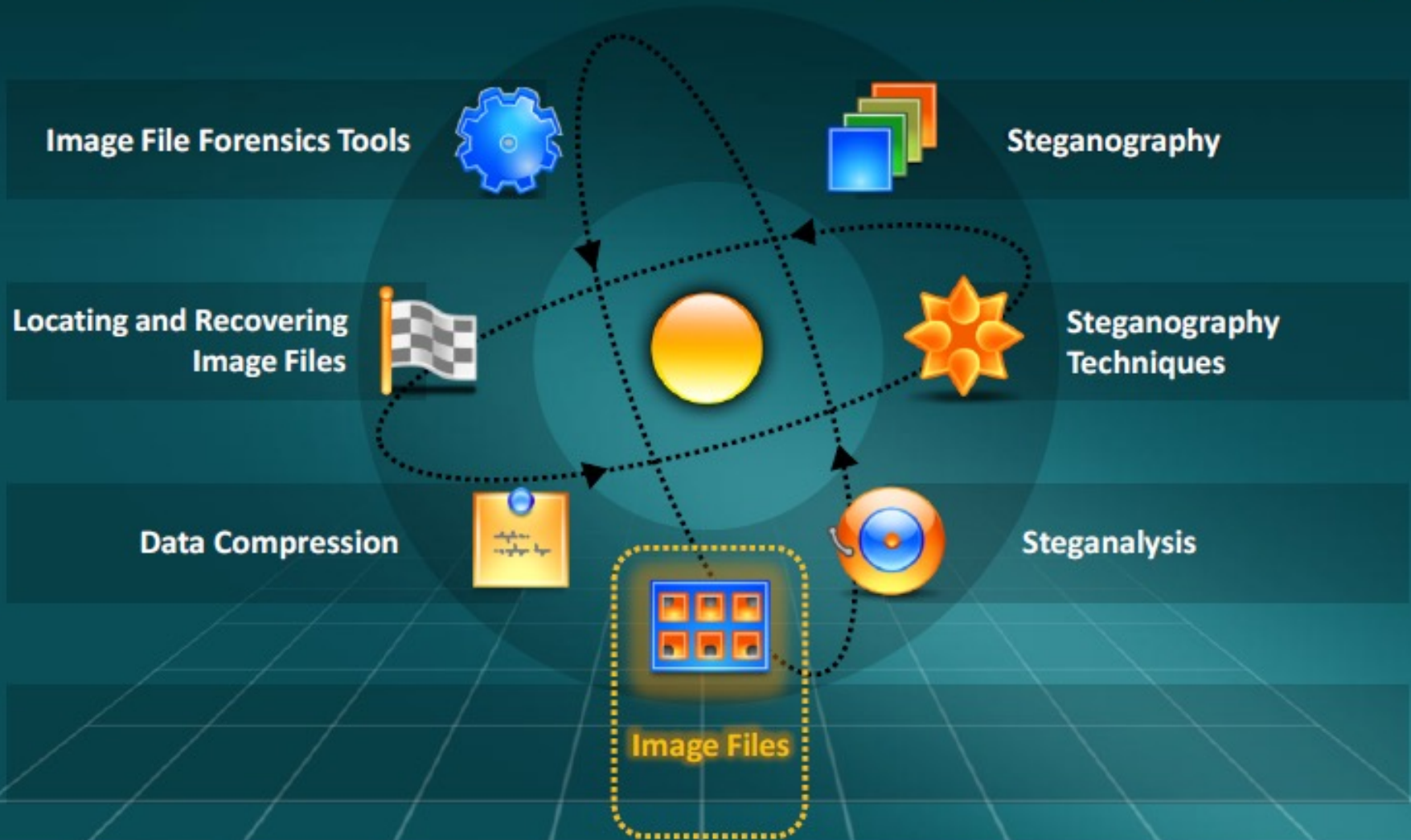
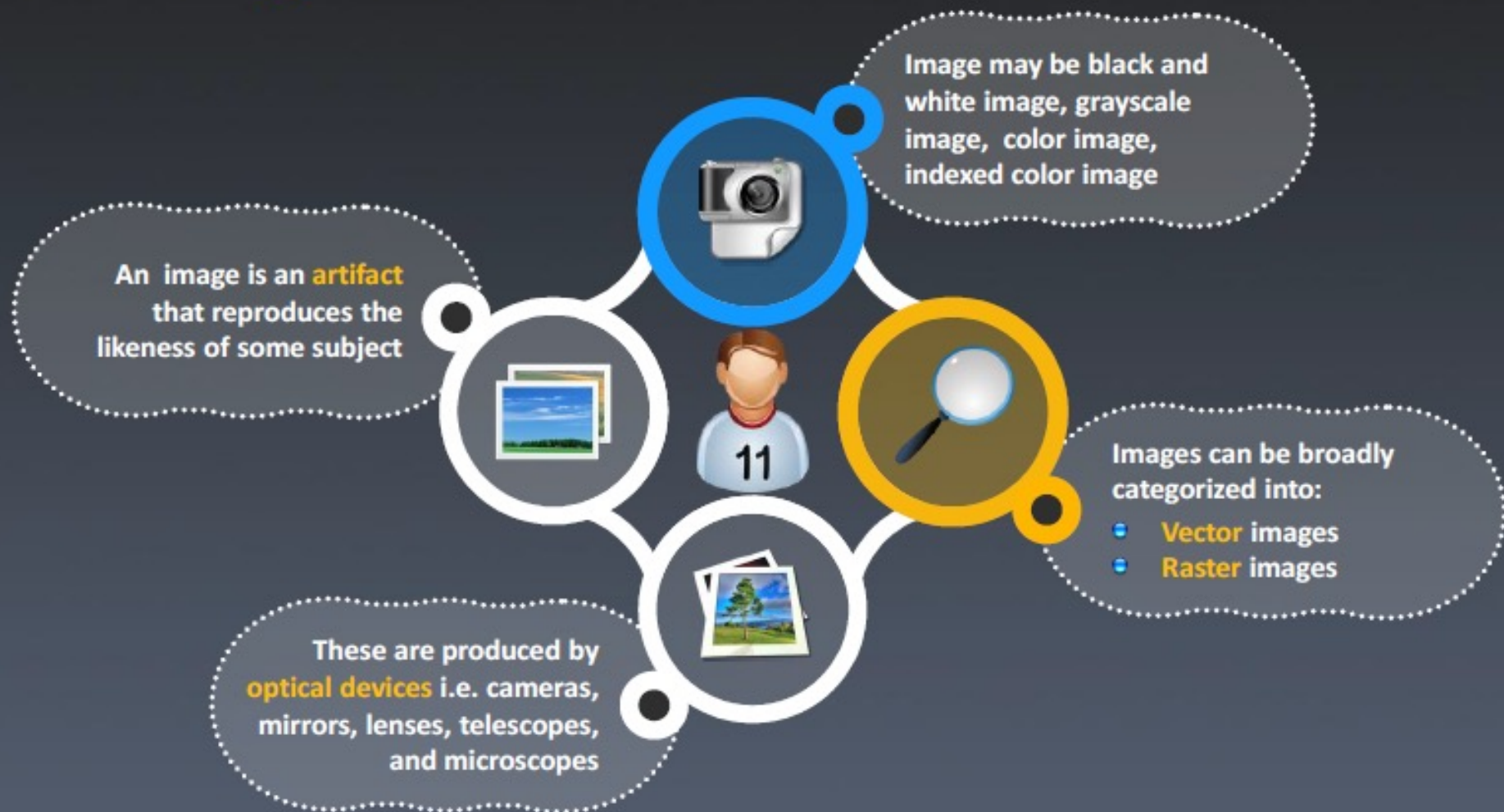


Image Files



Common Terminologies

Pixel

- Pixel (Picture Element) is a single point in a graphic image
- Number of pixels combine together to form an image

Bit Depth

- Refers to the number of colors available for each pixel in an image

Resolution

- Refers to the sharpness and clarity of an image
- The term describes monitors, printers, and bit-mapped graphic images

File Formats

- Particular way to encode information for storage in a computer file

Image File Size

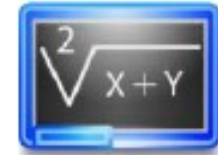
- Expressed as the number of bytes
- It increases with the bit depth of the pixel and number of pixels comprising an image

Compression

- Refers to the method of making image files smaller so that less disk space is used to store them

Understanding **Vector Images**

Vector graphics use **geometrical primitives** such as points, lines, curves, and polygons, which are all **based upon mathematical equations** to represent images in the computer



Smaller file size and useful for representing the images of various shapes



Can be indefinitely zoomed without loss in quality



Advantages



Applications such as moving, scaling, and rotating do not affect the quality of the image



Understanding Raster Images



A raster image is a **data file** or **structure** representing a generally rectangular grid of pixels, or points of color, on a computer monitor

Quality may be lost if raster graphics are **scaled** to a higher resolution

Color of each **pixel** is individually defined

Quality of a raster image is determined by the **total number of pixels** and the **amount of information** in each pixel

A colored raster image has pixels with **eight bits** of information for each of the red, green, and blue components



Metafile Graphics

Metafiles combine
raster and **vector**
graphics



Metafiles have
similar features of
both **bitmap** and
vector images



When metafiles are
enlarged, it results in
a loss of resolution,
giving the image a
shaded appearance



Understanding Image File Formats

- A file format is a particular way to **encode information** for storage in a computer file
- All image formats differ in their **ease of use**, the **size of the files** they produce, and their **image quality**

Filetype



Image File Formats



Standard Image File Formats

File Format	File Extension
Joint Photographic Experts Group (JPEG)	.jpg
JPEG 2000	.jp2
Graphics Interchange Format (GIF)	.gif
Tagged Image File Format (TIFF)	.tif
Windows Bitmap (BMP)	.bmp
Portable Network Graphics (PNG)	.png

Nonstandard Image File Formats

File Format	File Extension
Targa	.tga
Raster Transfer Language	.rti
Photoshop	.psd
Illustrator	.ai
Freehand	.h9
Scalable Vector Graphics	.svg
Paintbrush	.pcx

GIF (Graphics Interchange Format) (Cont'd)

- GIF is an **8-bit RGB bitmap image format** for images with up to 256 distinct colors per frame

Each color in the GIF color table is described in **RGB values**, with each value having a range of 0 to 255

It is a mechanism that makes images appear **faster** on-screen by first displaying a low-res version of the image and gradually showing the full version



This method is used to create the **illusion** of greater color depth by blending a smaller number of colored "dots" together

GIF supports **LZW lossless** compression algorithms

GIF (Cont'd)

1

Each file begins with a **Header** and a **Logical Screen Descriptor**



2

A **Global Color Table** may optionally be displayed after the Logical Screen Descriptor



3

Each image stored in the file contains a **Local Image Descriptor**, an optional **Local Color Table**, and a **block** of the image data

4

The last field in every GIF file is a **Terminator character**, which indicates the end of the GIF data stream

GIF

There are two versions of the GIF format:

GIF 87a

It supports LZW file compression, interlacing, 256-color palettes, and multiple image storage

GIF 89a

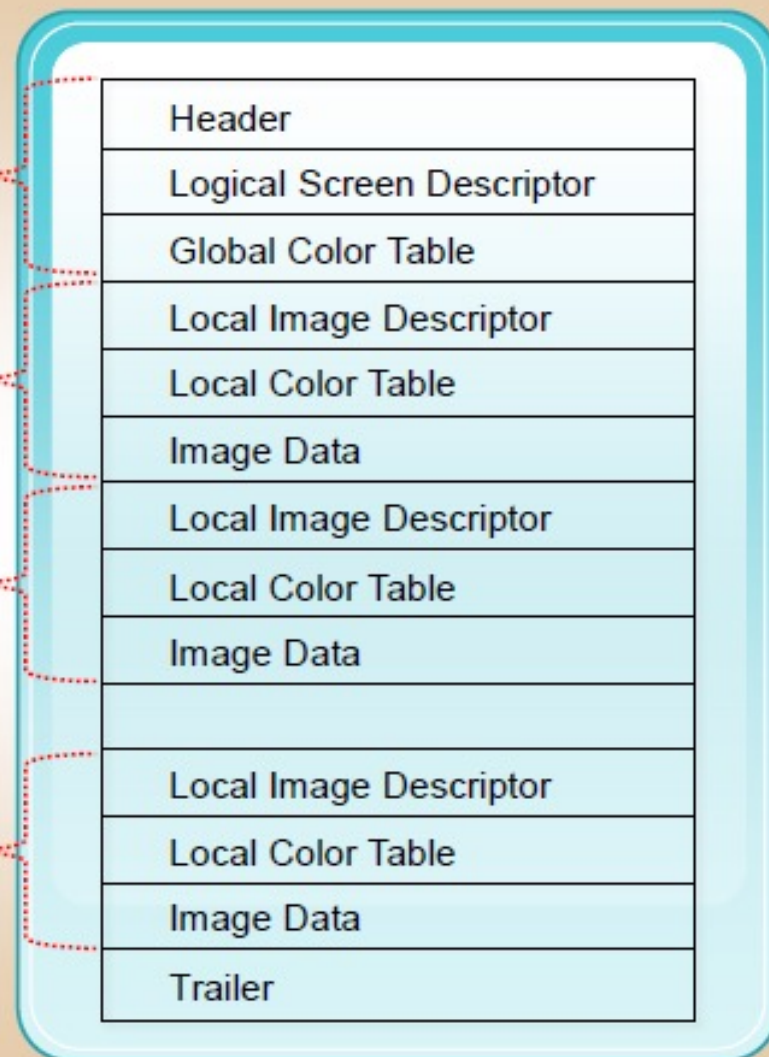
It supports properties such as background transparency, delay times, and image replacement parameters, which helps to store multiple images

Header and
Color Table
Information

Image 1

Image 2

Image n



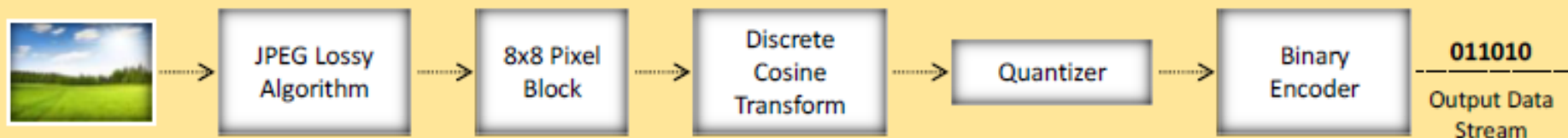
JPEG (Joint Photographic Experts Group)

JPEG is a commonly used method for compression of photographic images

It performs the file compression in four phases:



- 1 → The JPEG lossy algorithm divides the image in separate **blocks of 8x8 pixels**
- 2 → The compression algorithm then starts applying a **Discrete Cosine Transform (DCT)** for the whole block
- 3 → The compression algorithm **checks the JPEG image quality** requested by the user, and then computes the separate tables of quantization constants for luminance and chrominance
- 4 → Any of the Huffman or arithmetic encoding schemes are employed to **compress** these coefficients



JPEG File Structure (Cont'd)



JPEG Image

Contents	Name	Description
0xFF 0xD8	SOI	Start of image
Segments		
0xFF 0xD9	EOI	End of image



JPEG Segments

Description
Segment marker (2 bytes)
Segment size (2 bytes) excl. marker
Segment data

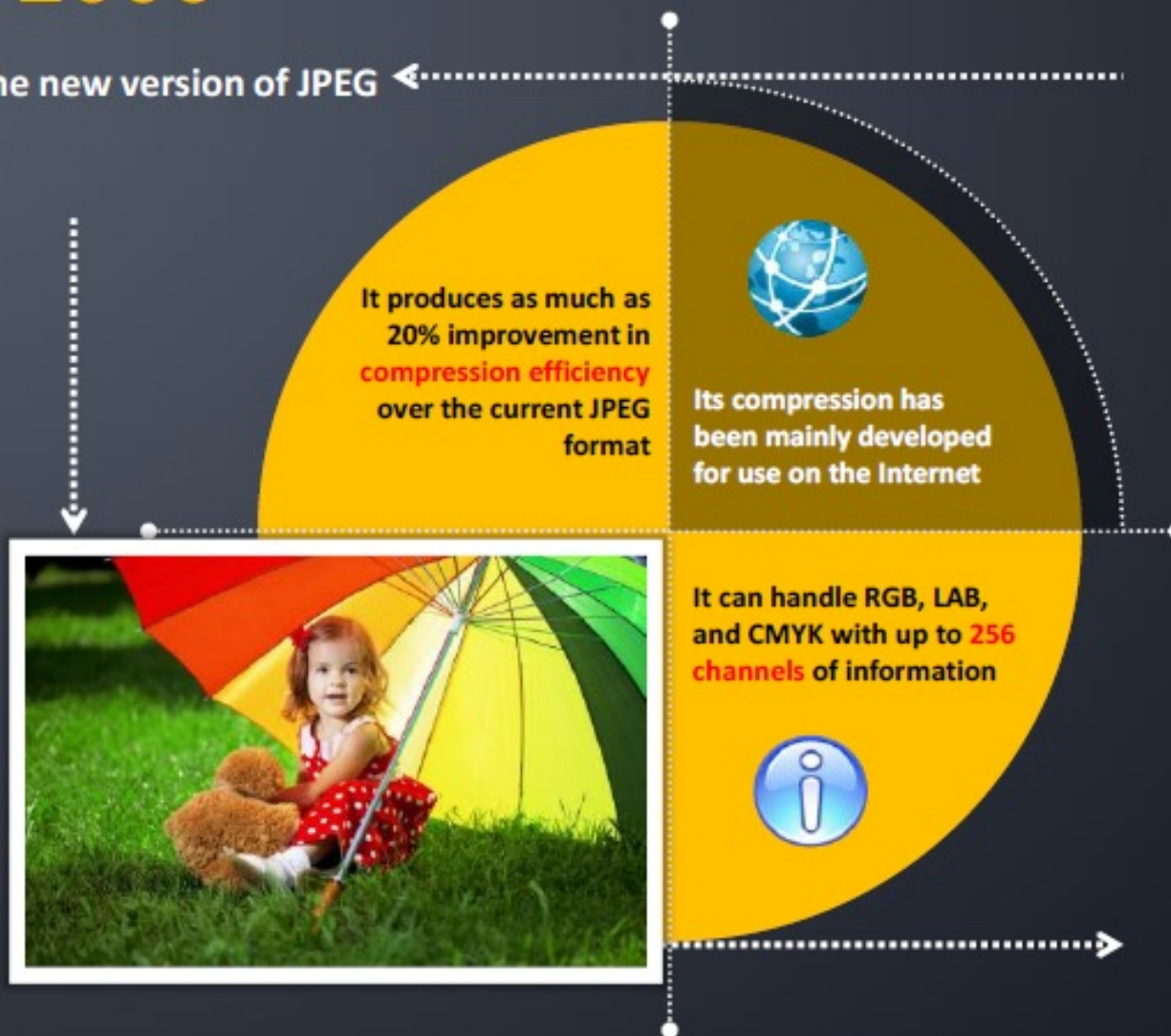
JPEG File Structure

Some JPEG Segment Markers

Contents	Name	Description
0xFF 0xE0	APP0	Application marker (in every JPEG file)
0xFF 0xDB	DQT	Quantization table
0xFF 0xC0	SOF0	Start of frame
0xFF 0xC4	DHT	Define Huffman Table
0xFF 0xDA	SOS	Start of scan
0xFF 0xED	APP14	This is the marker where Photoshop stores its information

JPEG 2000

JPEG 2000 is the new version of JPEG compression



BMP (Bitmap) File

- BMP is a standard file format for computers running the **Windows operating system**
- BMP images can range from **black** and **white** (1 bit per pixel) up to 24 bit color (16.7 million colors)
- Each bitmap file contains:

1. Header

Contains information about the **type**, **size**, and **layout** of a file



2. The RGBQUAD Array

Specifies the dimensions, compression **type**, and **color** format for the bitmap



3. Information Header

Color array contains a color table that is absent for bitmaps with 24 color bits because each pixel is represented by 24-bit **red-green-blue** (RGB) values in the actual bitmap



4. Image Data

These are the actual image data, represented by consecutive rows, or "scan lines," of the bitmap



BMP File Structure

Basic BMP File Format

Name		Size
Header		14 bytes
	Signature	2 bytes
	File Size	4 bytes
	reserved	4 bytes
	DataOffset	4 bytes
Color Table		4 * NumColors bytes
	Red	1 byte
	Green	1 byte
	Blue	1 byte
	reserved	1 byte
Repeated NumColors times		
Raster Data		Info ImageSize bytes

Basic BMP File Format (Cont'd)

Name		Size
InfoHeader		40 bytes
	Size	4 bytes
	Width	4 bytes
	Height	4 bytes
	Planes	2 bytes
	BitCount	2 bytes
	Compression	4 bytes
	ImageSize	4 bytes
	XpixelsPerM	4 bytes
	YpixelsPerM	4 bytes
	ColorsUsed	4 bytes
	ColorsImportant	4 bytes

PNG (Portable Network Graphics)

PNG bitmap image format uses **lossless data compression**

PNG was created to improve upon and **replace GIF** (Graphics Interchange Format) as an **image-file format**

PNG supports:

- Indexed / Palette-based images (24-bit RGB or 32-bit RGBA colors)
- Grayscale images (with or without alpha channel)
- Transparency (both normal and alpha channel)



PNG Image

PNG File Structure



PNG Signature

- A PNG file starts with an 8-byte signature that contains decimal values (137 80 78 71 13 10 26 10)
- This signature shows that the **remainder** of the file contains a single PNG image
- This image consists of a series of chunks starting with an **IHDR chunk** and ending with an **IEND chunk**

Chunk Layout

- It is a series of chunks, each of which gives some information about the image
- Each chunk has a header specifying the **size** and **type** of chunk
- Each chunk consists of four parts: **Length**, **Chunk Type**, **Chunk Data**, and **CRC** (Cyclic Redundancy Check)

TIFF (Tagged Image File Format)

- Tagged Image File Format is a **flexible** and **platform-independent** image file format
- It supports numerous image processing applications

Extendibility

- This is the ability to **add new image types** without invalidating the older types



Portability

- TIFF is **independent of the hardware platform** and the operating system on which it executes



Revisability

- TIFF was designed to be an efficient medium for **exchanging image information**
- It is used as a native internal data format for image editing applications



TIFF File Structure (Cont'd)

TIFF files are made up of three unique data structures:

Image File Header (IFH)

- The IFH is an **8-byte structure** located at offset zero in the file
- The IFH contains important information necessary to correctly interpret the **remainder** of the TIFF file



Image File Directory (IFD)

- An IFD consists of a count N, the number of directory entries
- Each entry is of **12 bytes**
- If more than one IFD is present, the file contains more than one image

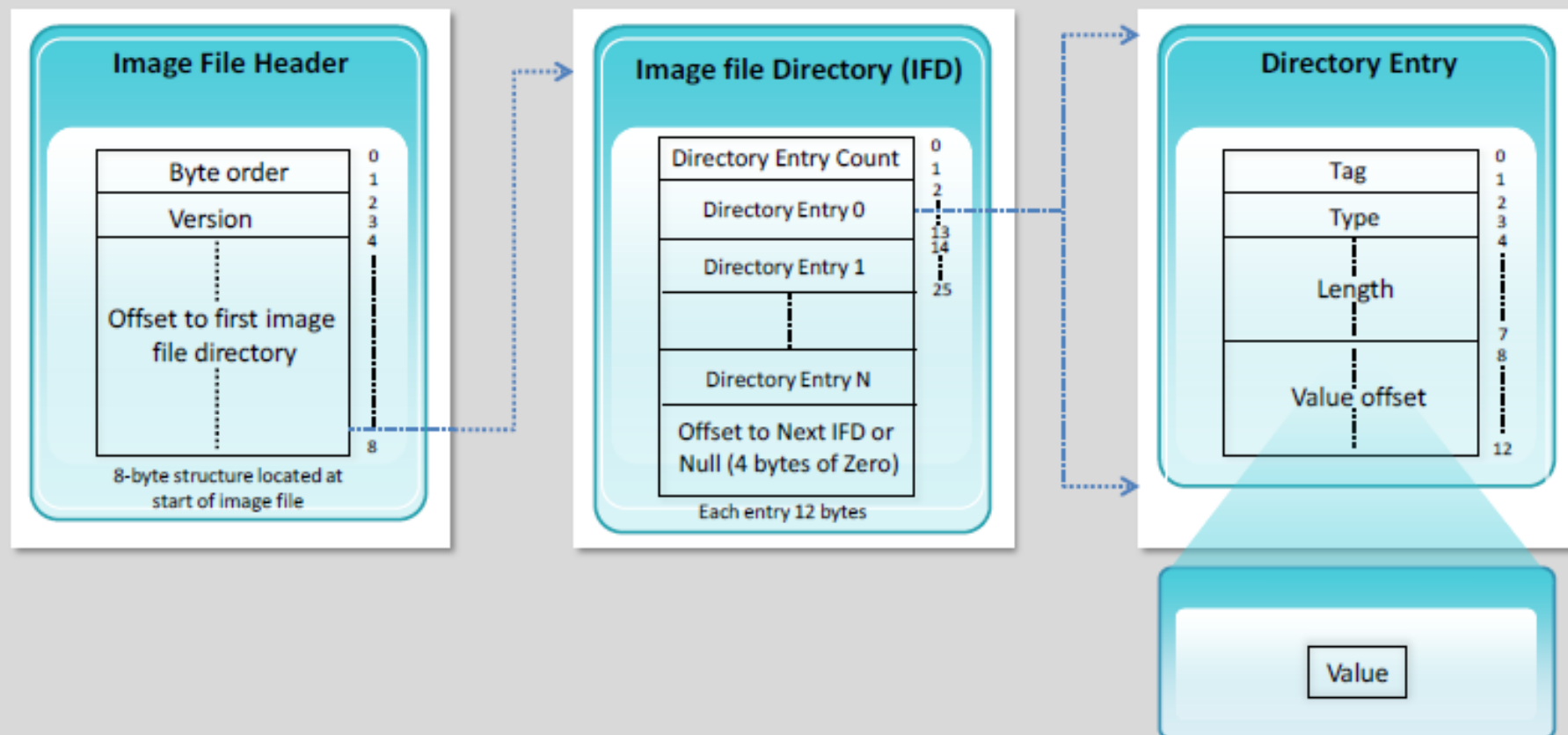


Directory Entry (DE)

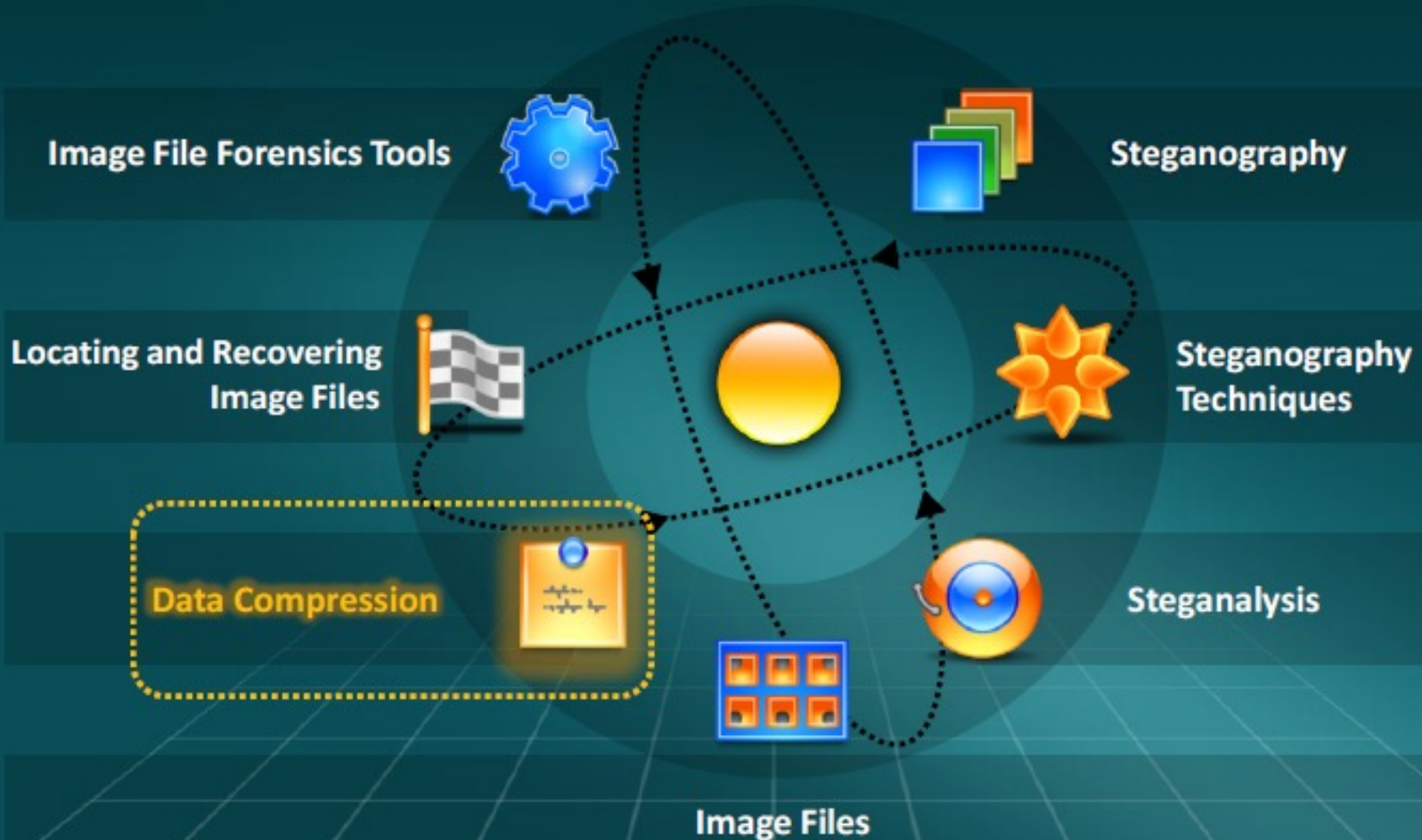
- Each DE is **exactly 12 bytes** in length
- It is segmented into four fields



TIFF File Structure



Module Flow



Understanding Data Compression



Data compression means **encoding the data** to take up less storage space and less bandwidth for **transmission**



It helps in **saving cost** and high **data manipulation** in many **business applications**



There are two techniques of data compression:



Lossless Compression, which maintains the data integrity



Lossy Compression, which does not maintain the data integrity

How Does **File Compression** Work?

In John F. Kennedy's 1961 inaugural address, he delivered this famous line:
"Ask not what your country can do for you -- ask what you can do for your country"



When you go through Kennedy's famous words, pick out the words that are repeated and put them into the numbered index

So, if this is your dictionary:

- ask
- what
- your
- country
- can
- do
- for
- you

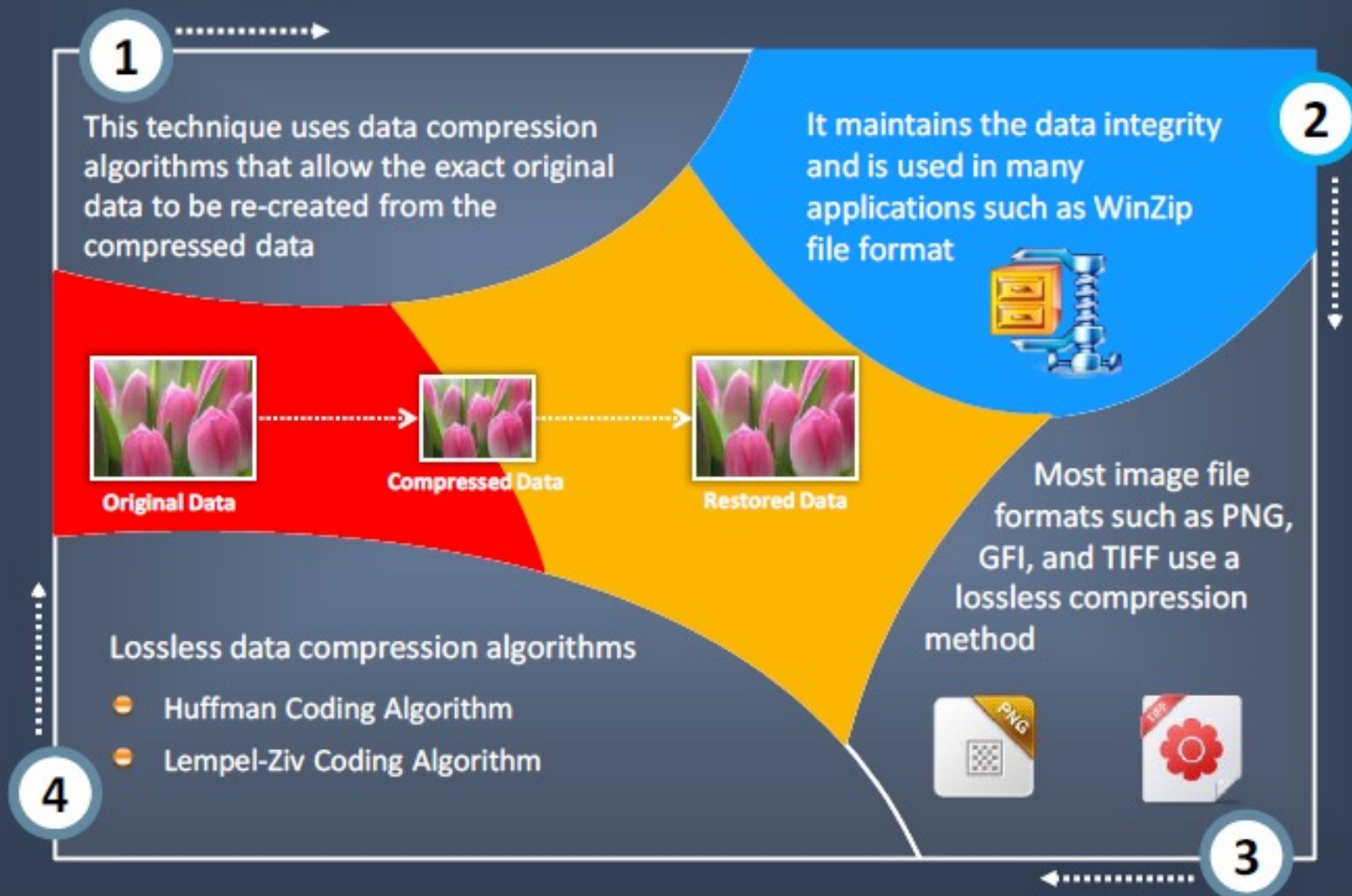


The sentence now reads:

"1 not 2 3 4 5 6 7 8 -- 1 2 8 5 6 7 3 4"

Then, simply write the number instead of writing out the whole word

Lossless Compression



Huffman Coding Algorithm (Cont'd)



Huffman Coding Algorithm is a **fixed-to-variable length code**

A Huffman encoder takes a block of input characters with a **fixed length** and produces a block of output bits of **variable length**

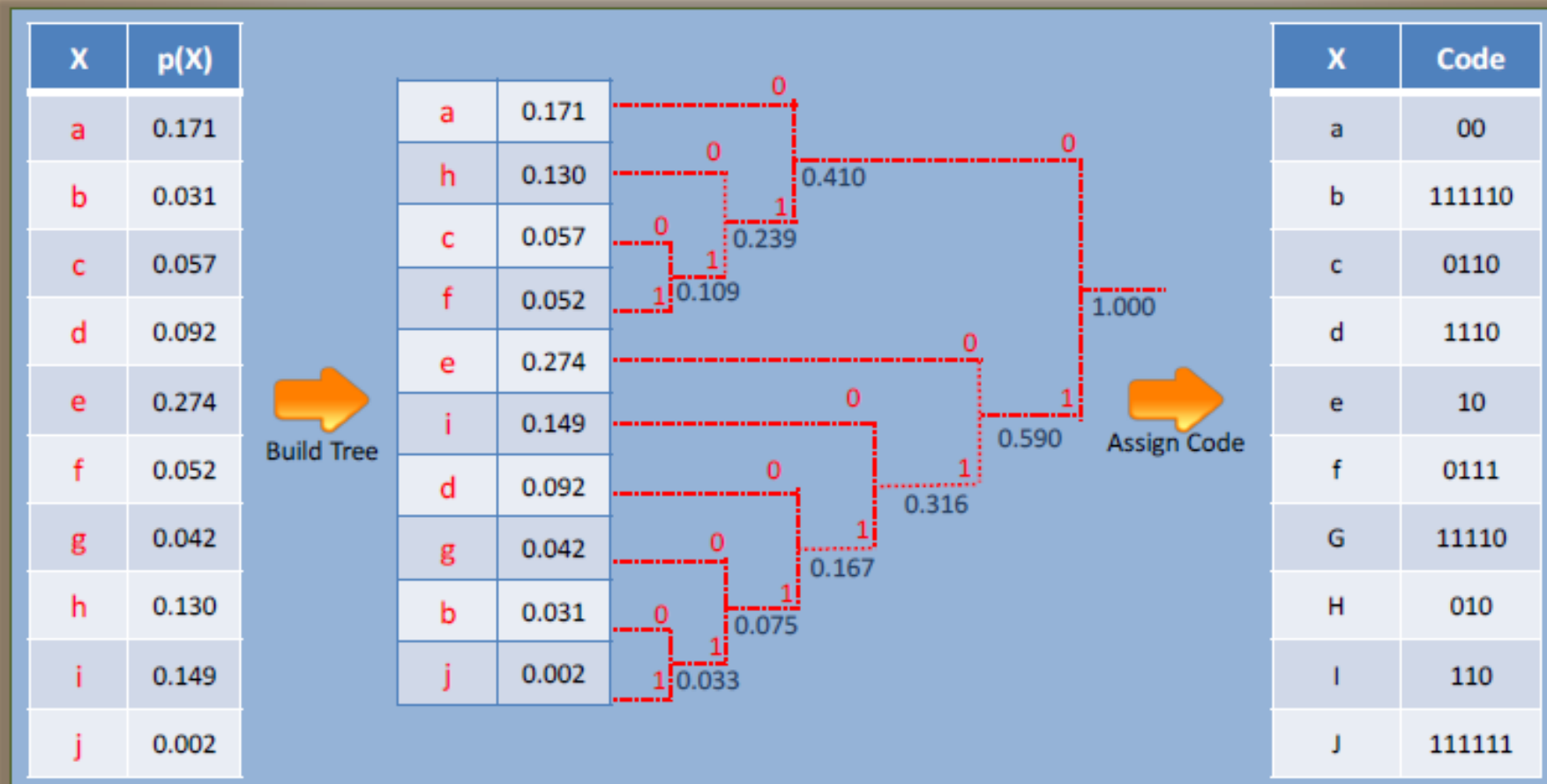
The basic idea in Huffman coding is to assign a **short codeword** to those input blocks with **high probabilities** and a **long codeword** to those with **low probabilities**

A Huffman code is designed by **merging** the two least probable **characters** together

Repeat this merging **process** until there is only **one character remaining**

Huffman Coding Algorithm

Example shows how it works:



Lempel-Ziv Coding Algorithm (Cont'd)

Lempel-Ziv Coding Algorithm is a variable-to-fixed length code



In this, the input sequence is parsed into non-overlapping blocks of different lengths

The Lempel-Ziv code is not designed for any particular source but for a large class of sources



Dictionary of these blocks is constructed and the following algorithm is used

Encoding Algorithm

- Initialize the dictionary to contain all blocks of length **one** ($D=\{a,b\}$)
- Search for the **longest block W** in the dictionary

- Encode **W** by its **index** in the dictionary
- Add **W** followed by the **first symbol** of the next block to the dictionary
- Go to Step 2



Lempel-Ziv Coding Algorithm

An example of encoding:

Data a b b a a b b a a b a a b b a

0 1 1 0 2 4 2 6 5 5 7 3 0

Dictionary

Index	Entry	Index	Entry
0	a	7	b a a
1	b	8	a b a
2	a b	9	a b b a
3	b b	10	a a a
4	b a	11	a a b
5	a a	12	b a a b
6	a b b	13	b b a



Lossy Compression

- Lossy methods provide a **high degree of compression** and small compressed files, but during decompression, a certain amount of data is lost
- It does not maintain **data integrity**
- It is never used for **business data** or **text files**



Original Data



Compressed Data



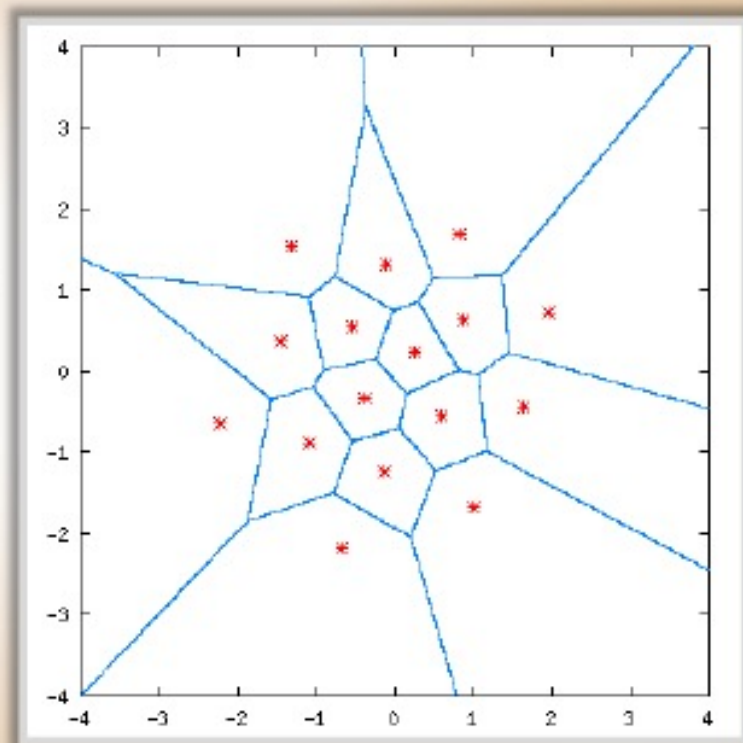
Restored Data

Vector Quantization

Vector quantization is a lossy **data compression** technique



This technique is based on the principle of **block coding**, which means it replaces a block of information with an approximate **average value**



Module Flow



Best Practices for Forensic Image Analysis



Forensic Image Processing Using **MATLAB**



1

It ensures the **image processing steps** used are completely **documented** and hence can be replicated



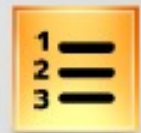
2

The **source code** for all image processing functions is accessible for scrutiny and testing



3

It ensures that **numerical precision is maintained** all the way through the enhancement process



4

Advanced **image processing algorithms** are used



Advantages of MATLAB

Recording of the processing used

- MATLAB is used to **process images** by writing function files or script files
- These files form a **formal record** of the processing used and ensure that the final results can be **tested and replicated**

Access to implementation details

- Functions written in the MATLAB language are **publicly readable** as plain text files

Advantages



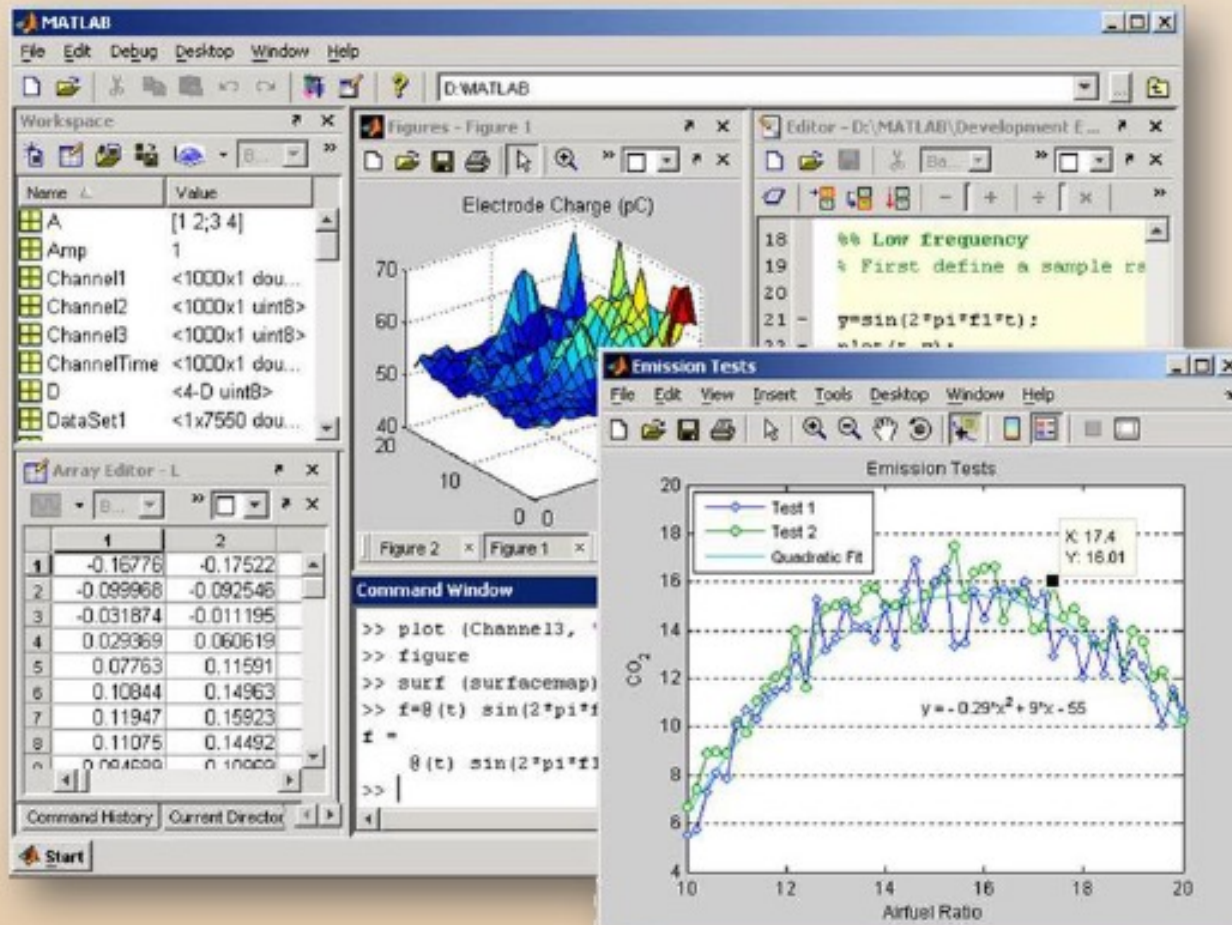
Numerical accuracy

- It ensures maximal **numerical precision** in the final result
- An image can be read into memory and the data cast into double precision **floating point values**

Advanced algorithms

- It provides strong **mathematical and numerical support** for the implementation of advanced algorithms

MATLAB Screenshot



<http://www.mathworks.com>

Locating and Recovering Image Files



Salvaging

- Collecting and **regenerating the image** from pieces of an image file dispersed into many areas on the disk is known as salvaging



Regenerating
Image



Carving

- It is the process of **data recovery**
- It uses the database of headers and footers (essential string of bytes) for a specific file type and **recovers files from the raw disk image**
- File carving also works if the file system **metadata** has been **destroyed**



Analyzing Image File Headers



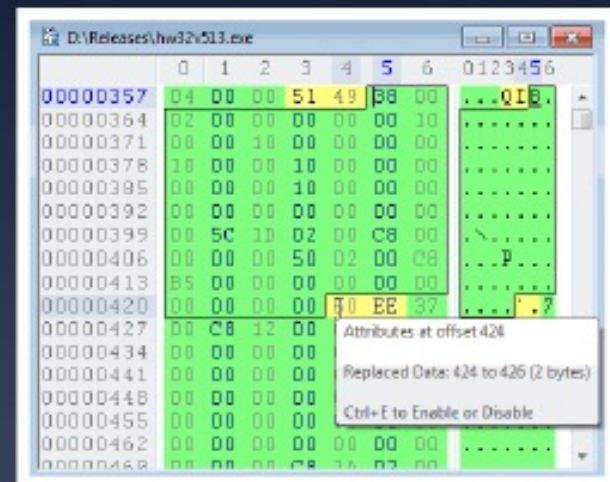
Investigators **analyze image file headers** when new file extensions are present that forensic tools cannot recognize



File headers are accessed with the help of a **hexadecimal editor** such as the **Hex Workshop**



Hexadecimal values present in the header can be used to **define a file type**



Repairing **Damaged** Headers (Cont'd)

JPEG files would include letters
"JFIF" after hexadecimal values

Example: JPEG files have a hexadecimal value of:
FF D8 FF E0 00 10



The **HEX Workshop** application can
be used to repair the damaged
headers by the process of
comparison



Investigators recover data
remnants from **free space**



This data would be **similar to
headers** from common
image files



Header data that is partly
overwritten can be used to
repair the damaged headers

Repairing **Damaged** Headers



Hex Workshop - D:\Releases\hw32v513.exe

File Edit Disk Options Tools Plug-Ins Window Help

00000357 04 00 00 51 49 B8 00 ...Q1E

00000364 02 00 00 00 00 00 10 ...

00000371 00 00 10 00 00 00 00 ...

00000378 10 00 00 10 00 00 00 ...

00000385 00 00 00 10 00 00 00 ...

00000392 00 00 00 00 00 00 00 ...

00000399 00 5C 10 02 00 C8 00 ...

00000406 00 00 00 50 02 00 C8 ...P...

00000413 B5 00 00 00 00 00 00 ...

00000420 00 00 00 00 F0 EE 37 ...R..?

00000427 00 C8 12 00 00 00 00 ...

00000434 00 00 00 00 00 00 00 ...

00000441 00 00 00 00 00 00 00 ...

00000448 00 00 00 00 00 00 00 ...

00000455 00 00 00 00 00 00 00 ...

00000462 00 00 00 00 00 00 00 ...

00000469 00 00 00 00 00 00 00 ...

Attributes at offset 424
Replaced Data: 424 to 426 (2 bytes)
Ctrl+E to Enable or Disable

hw32v514... hw32v513...

D:\Releases\hw32v513.exe vs D:\Releases\hw32v514.exe

Type	Source	Count	Count	Target	Count	Count
Matched	00000000	360	0168	00000000	360	0168
Replaced	00000360	2	02	00000360	2	02
Matched	00000362	62	3E	00000362	62	3E
Replaced	00000424	2	02	00000424	2	02
Matched	00000426	183142	0002CB66	00000426	183142	0002CB66
Replaced	00182568	11	02	00182568	11	02

Compare Checksum Find Bookmarks Output

Cursor: 424 Caret: 362 3670312 bytes CVR MOD READ

Reconstructing File Fragments

- **Corruption of the data** prevents investigators from reconstructing file fragments for image files

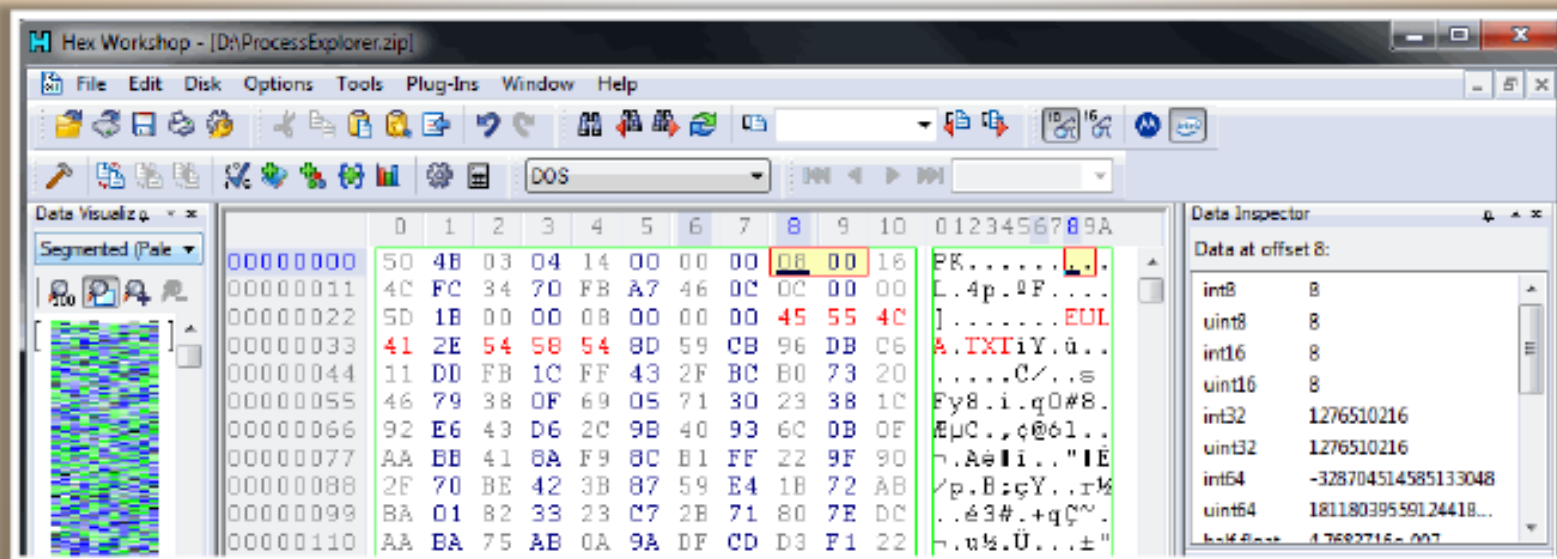
- Data corruption can be:

- Accidental
- Intentional



- File fragments can be reconstructed by **examining a suspect disk** with the help of the DriveSpy application

- Investigators can build the case based on the data reconstructed



Identifying Unknown File Formats

To understand unknown image file formats ,you should know about non-standard file formats:

- Targa (.tga)
- Raster Transfer Language (.rtf)
- Photoshop (.psd)
- Illustrator (.ai)
- Freehand (.h9)
- Scalable vector graphics (.svg)
- Paintbrush (.pcx)



Tools to identify the unknown file formats:

- IrfanView
- ACDSee Photo Manager 12
- Thumbsplus
- AD Picture Viewer Lite
- Max
- FastStone Image Viewer
- XnView



Identifying **Image File** Fragments



The first step in recovering the deleted data files is to **identify the image files' fragments**



Recover all the fragments to re-create the image if the image file is fragmented across different disk areas



Recovering a piece of file is called **salvaging** or **carving**



After recovering the parts of the fragmented image file, **restore the fragments** and continue the forensic investigation

Identifying Copyright Issues on Graphics



Section 106 of the 1976 Copyright Act:

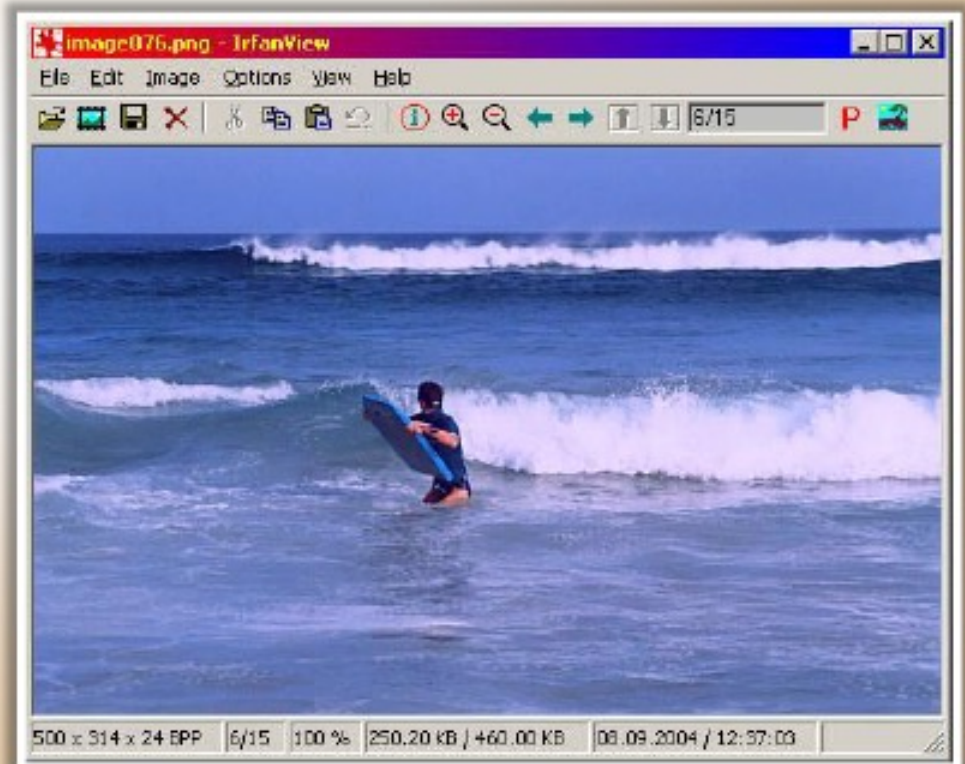
The owner of copyright under this title has the exclusive rights to do and to authorize any of the following:

- To reproduce the **copyrighted work** in copies or phonorecords;
- To prepare **derivative works** based upon the copyrighted work;
- To **distribute** copies or phonorecords of the copyrighted work to the public by sale or other transfer of ownership, or by rental, lease, or lending;
- In the case of literary, musical, dramatic, and choreographic works, pantomimes, and motion pictures and other audiovisual works, to perform the copyrighted work publicly;
- In the case of literary, musical, dramatic, and choreographic works, pantomimes, and pictorial, graphic, or sculptural works, including the individual images of a motion picture or other audiovisual work, to display the copyrighted work publicly;
- In the case of sound recordings, to perform the copyrighted work publicly by means of a **digital audio transmission**

Picture Viewer: **IrfanView**

■ IrfanView is an **image/graphics** viewing program that supports many file formats, such as:

- Targa (.tga)
- Illustrator (.ai)
- Scalable vector graphics (.svg)
- FlashPix (fpx)
- Portable network graphics (.png)
- Joint photographic experts group (.jpg)

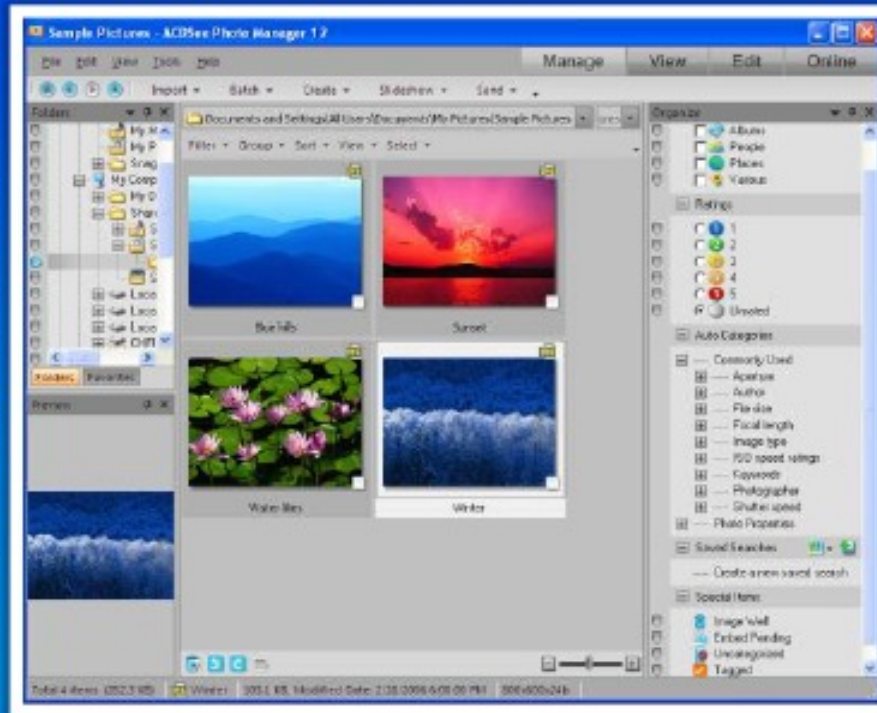


<http://www.irfanview.com>

Picture Viewer: **ACDSee Photo Manager 12**

ACDSee Photo Manager 12 is an **image viewing program** that enables investigators to:

- 01** Find images
- 02** View images
- 03** Manage image files on the drive
- 04** Search and view unknown file formats



<http://store.acdsee.com>

Picture Viewer: **Thumbsplus**

- ThumbsPlus is an **image viewing program** that enables investigators to:
 - View images from a drive **database**
 - View files other than images such as **audio** and **multimedia** files
 - **Catalog** image files for future reference
 - Save **RAW** files



<http://www.cerious.com>

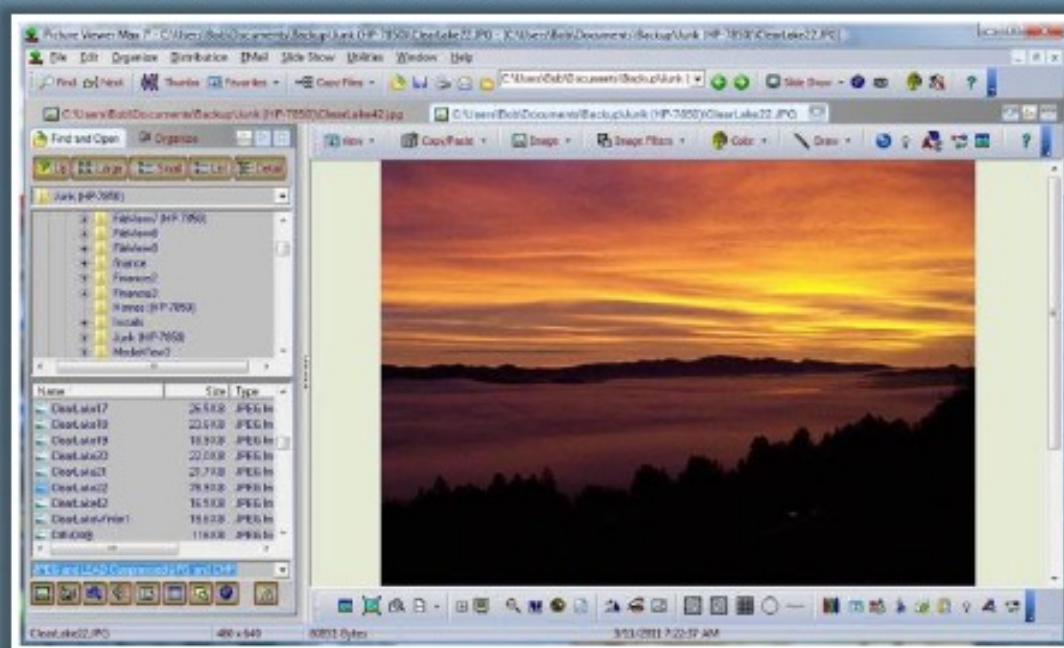
Picture Viewer: AD Picture Viewer Lite

- AD Picture Viewer Lite is an image viewer that support all popular **image file formats**
- It allows you to **view, print, organize,** and **catalog** the image
- It opens all images in a folder, including in subfolders, using a **single command**



Picture Viewer Max

Picture Viewer Max is an **image** and **multimedia** viewer for Windows 7 that helps to locate, view, edit, print, organize, and send/receive picture/image files over the Internet



<http://accessoryware.com>

Picture Viewer: **FastStone Image Viewer**



<http://www.faststone.org>

- FastStone Image Viewer is an **image browser, converter, and editor**
- It has many features that include image viewing, management, comparison, red-eye removal, emailing, resizing, cropping, retouching and color adjustments
- Features:
 - Supports **common image formats**, loading and saving of JPEG, JPEG2000, GIF, BMP, PNG, PCX, TIFF, WMF, ICO, CUR, and TGA
 - Supports **zoom - full screen viewer**
 - Crystal **clear** and **customizable** magnifier
 - Image **EXIF metadata** support

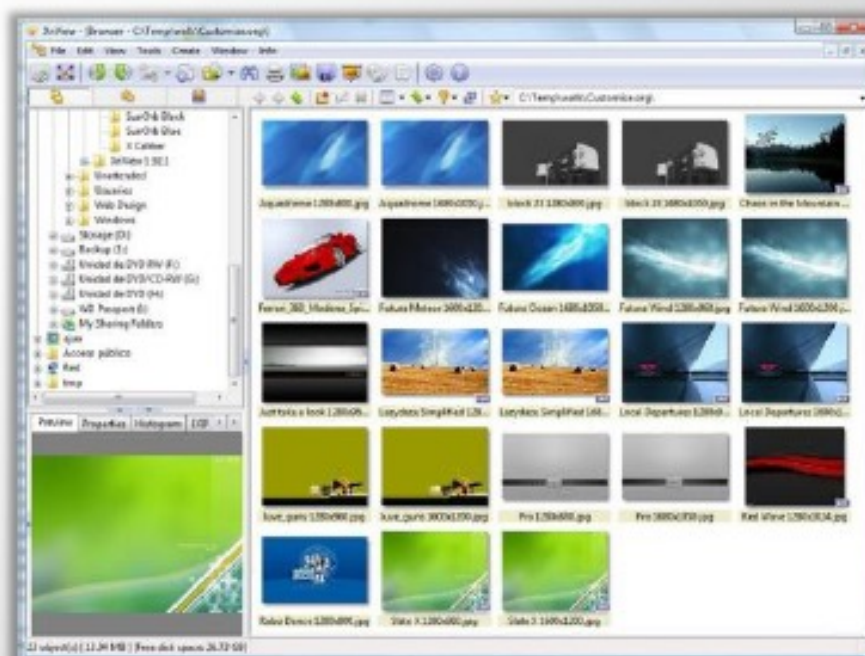
Picture Viewer: **XnView**

- XnView is software for **viewing** and **converting** graphic files
- It exists for Windows, MacOS X, Linux x86, Linux ppc, FreeBSD x86, OpenBSD x86, NetBSD x86, Solaris sparc, Solaris x86, Irix mips, HP-UX, and AIX



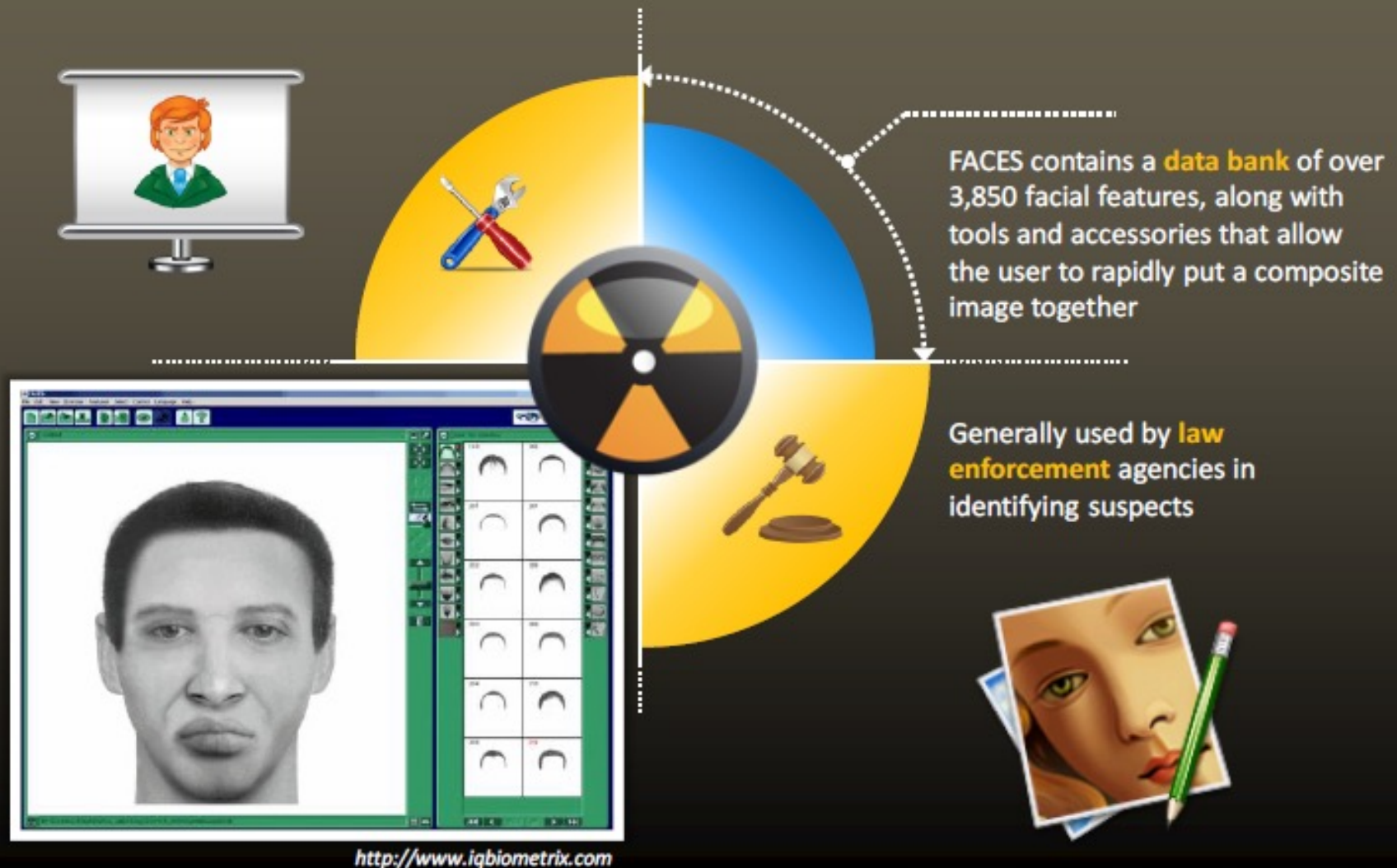
Features

- Imports about **400** graphic file formats
- Exports about **50** graphic file formats
- Supports multipage TIFF, animated GIF, and animated ICO
- Supports Image IPTC, EXIF metadata
- Supports lossless rotate and crop (jpeg)
- Creates **web pages**



<http://www.xnview.com>

Faces - Sketch Software



Digital Camera Data Discovery Software: **File Hound**

File Hound is a software package that helps to deal with crimes involving **digital pictures**

It searches images based on **file signature**



It can preview/print **reports** based on search results



It **distinguishes** PNG, GIF, JPG, TIF, WMF, BMP, ICO



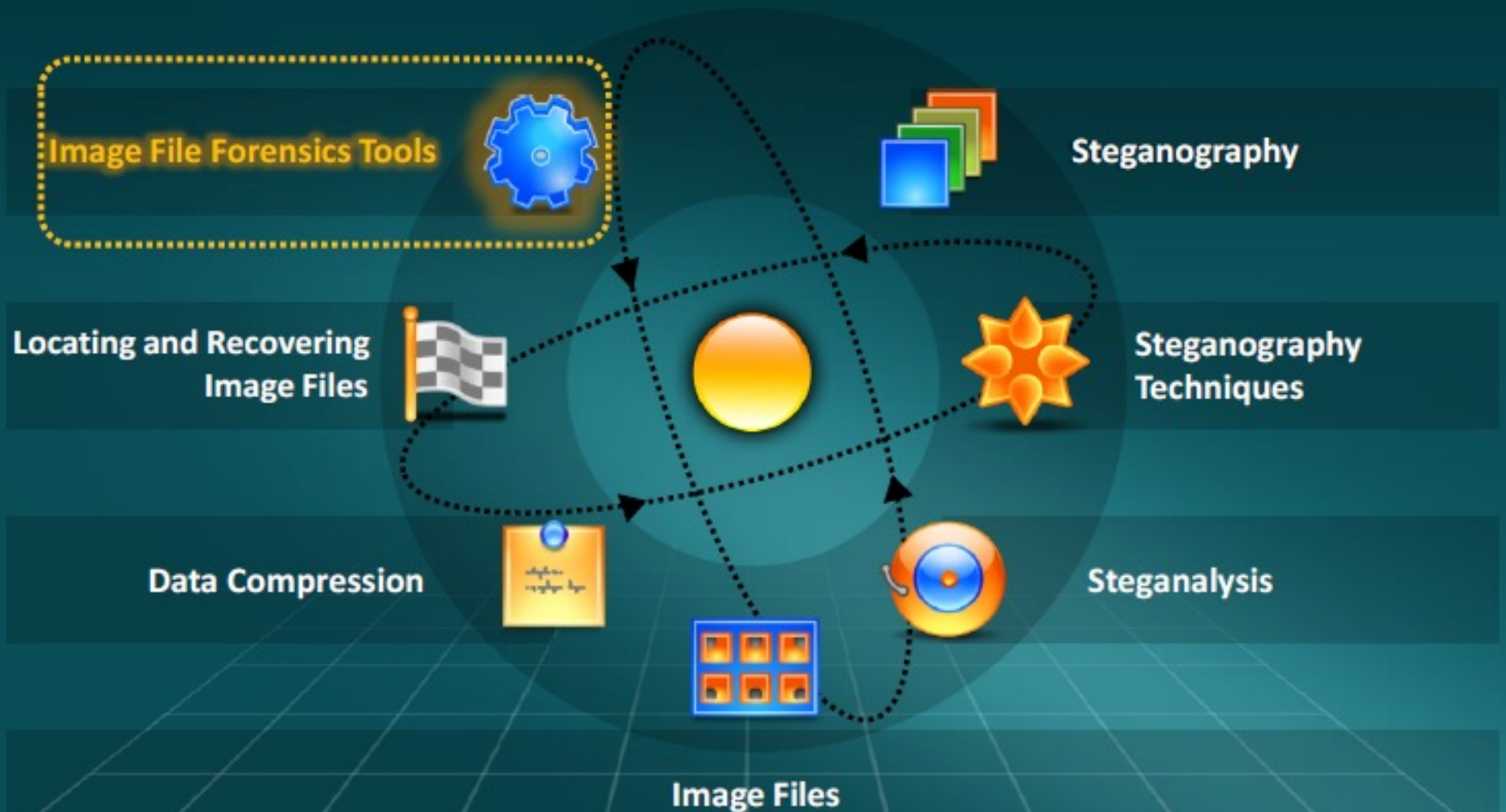
It **previews** thumbnail images and file information



It searches for files based on **filenames**

<http://filehound.cerias.purdue.edu>

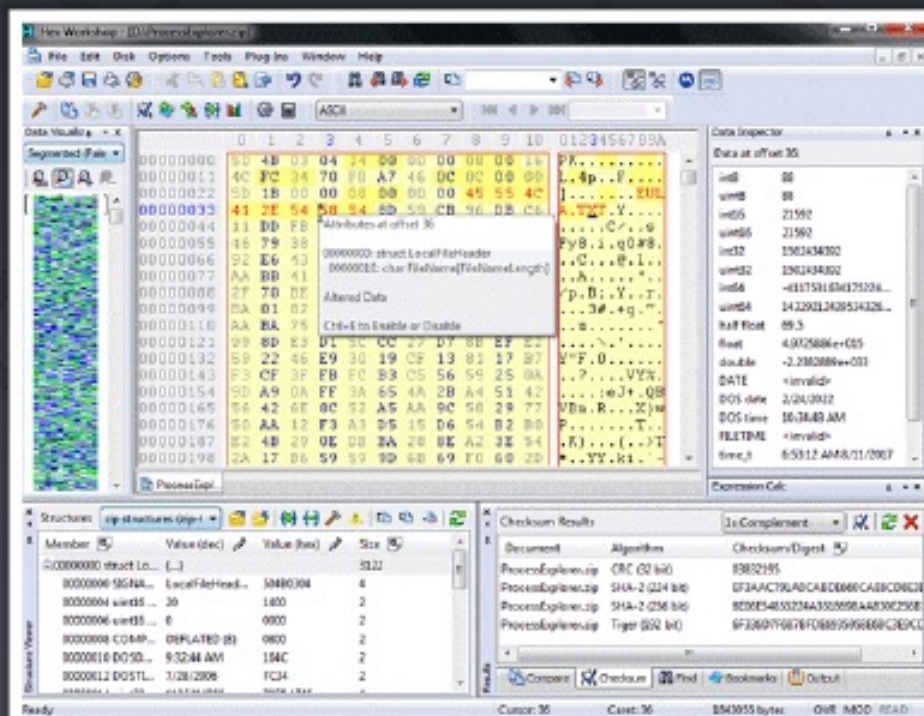
Module Flow



Hex Workshop

- The Hex Workshop application can **detect** and **write messages** onto a file
- Investigators use the Hex Workshop tool to **reconstruct** the damaged file headers

EB F6
52 00
90 80
4E F6



GFE Stealth™ - Forensics Graphics File Extractor

- GFE Stealth™ is used to **identify** and **capture graphics files** (pictures) stored on computer hard disk drives
- It is a computer forensics tool that helps identify whether or not a **targeted** computer system contains **evidence** in the form of graphics files

Used effectively in computer investigations involving the **distribution of child pornography** and **identity theft**

Used in the **recovery** of **deleted** graphics files from computer **hard disk drives** and/or digital **flash memory chips**

Used to find evidence in **corporate, civil** and **criminal investigations** that involve computer graphics files

Used with other computer forensic software to quickly **reconstruct** and **view** previously **deleted BMP, GIF** and **JPEG** graphics files stored on computer **storage media**

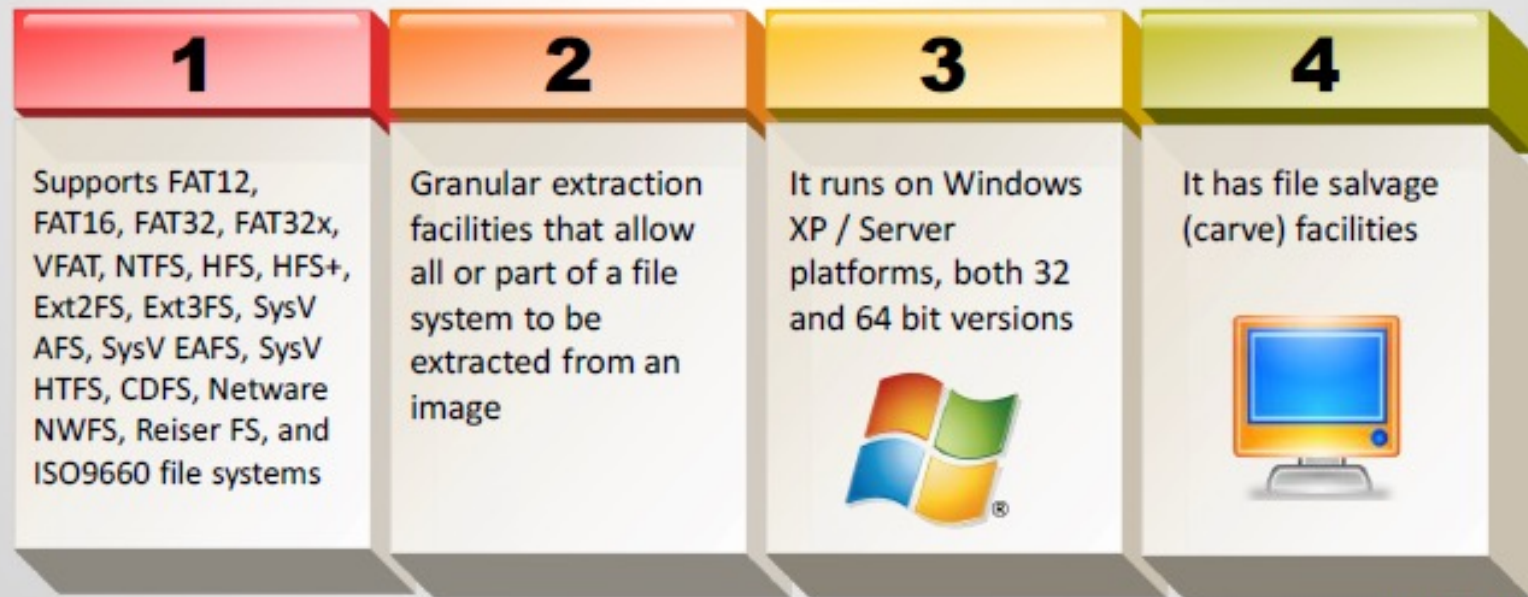
<http://www.forensics-intl.com>



Ilook

- Ilook is a multi-threaded, Unicode compliant, fast, and efficient forensic analysis tool designed to **analyze** images taken from **seized computer systems** and other **digital media**
- It can be used to examine images obtained from other forensic imaging tools that produce a **raw bit stream image**

Features



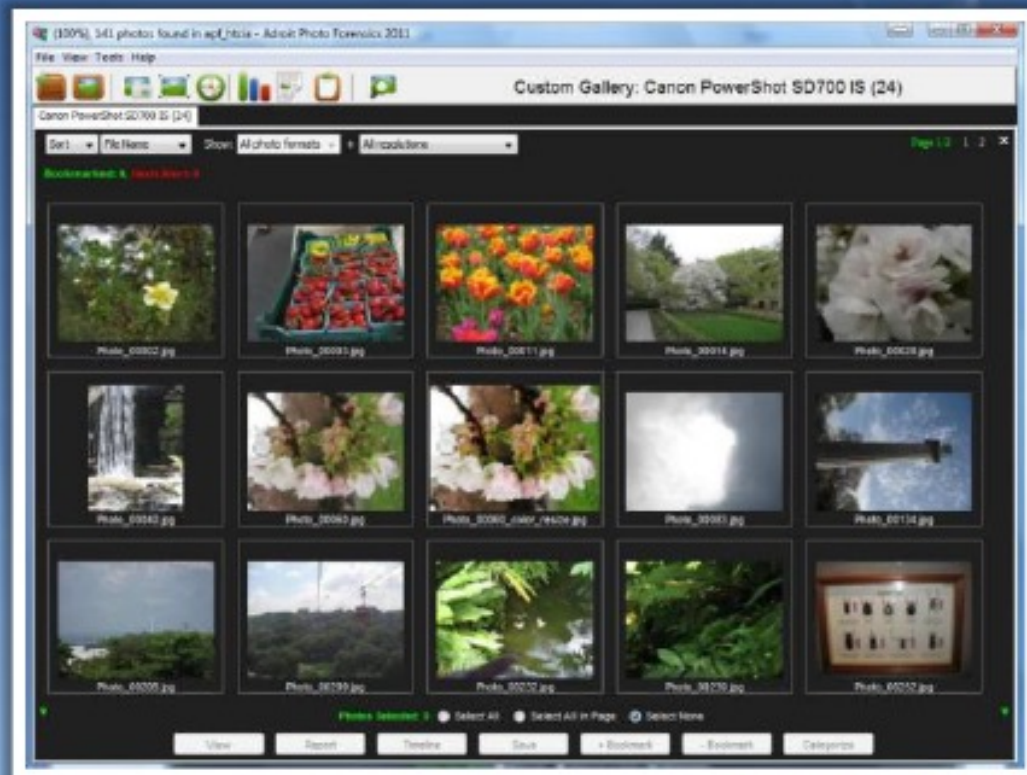
<http://www.perlustro.com>

Adroit Photo Forensics 2011

Adroit Photo Forensics 2011 is designed for professional evidence recovery and uses **SmartCarving** and **GuidedCarving** technologies for complete recovery of photo evidence

Features:

- Automatically generate case details based on selected evidence
- Supports Encase disk images (single and split), RAW/DD images (single and split), and logical/physical drives
- Generate MD5 and SHA1/SHA256 hashes of photos recovered
- Photo specific grouping based on camera, EXIF Date stamps etc.
- Carving support for unallocated space and slack space between partitions



<http://digital-assembly.com>

Digital Photo Recovery

1

Digital Photo Recovery tool recovers images from the camera's **memory card**

2

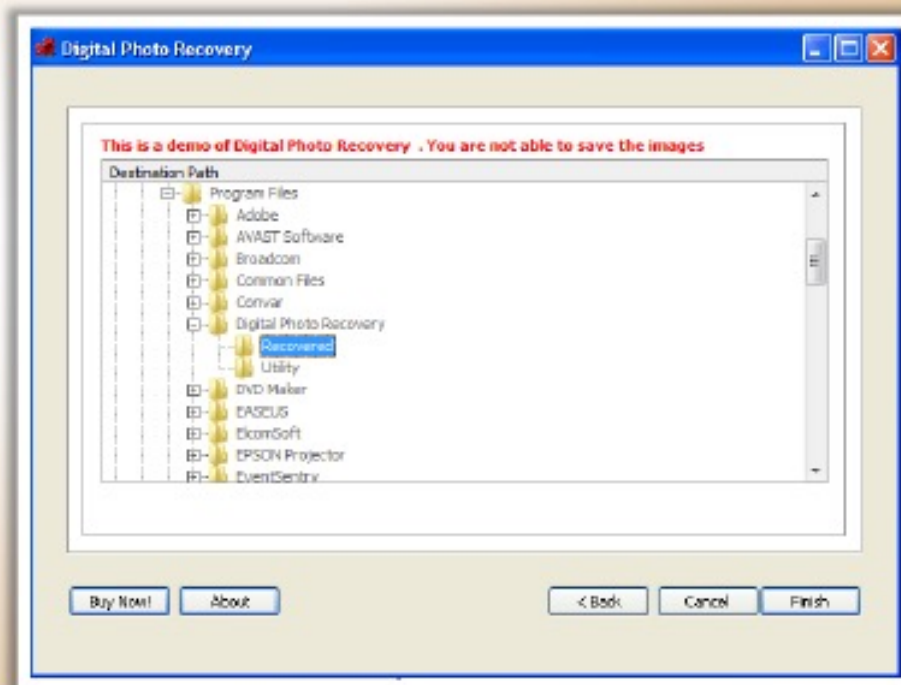
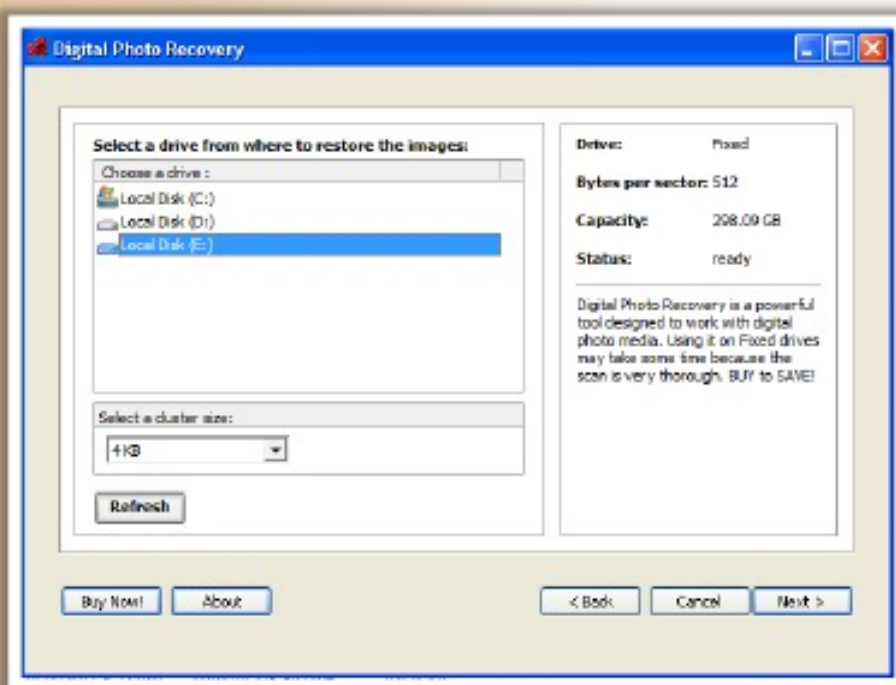
It recovers **lost, deleted, and formatted** digital photos from all media types

3

It recovers lost or deleted digital photos from:

- Hard drive
- Secure Digital Card (SD card)
- MultiMediaCard (MMC)
- CompactFlash (CF)
- Floppy Disk
- USB Memory Card
- Digital Cell Phones
- DVD+R DVD-R DVD+RW DVD-RW
- PDA PC Card

Digital Photo Recovery Screenshots



<http://www.photosrecovery.com>

Stellar Phoenix Photo Recovery Software

Stellar Phoenix Photo Recovery Software recovers lost/deleted/formatted **photos**, **music**, and **video** files from Windows based systems



It recovers photos lost due to **deletion**, **corruption** or **formatting** of SD cards and hard drives



It **recovers** photos from digital cameras, external hard drives, laptops and desktops, CDs, DVDs, USB Drives, and memory cards

Zero Assumption Recovery (ZAR)

- Zero Assumption Recovery (ZAR) software recovers **digital images** from memory cards of digital cameras
- It supports various file formats such as GIF, JPEG, TIFF, CRW (Canon RAW data), MOV, AVI movie, WAV, etc.

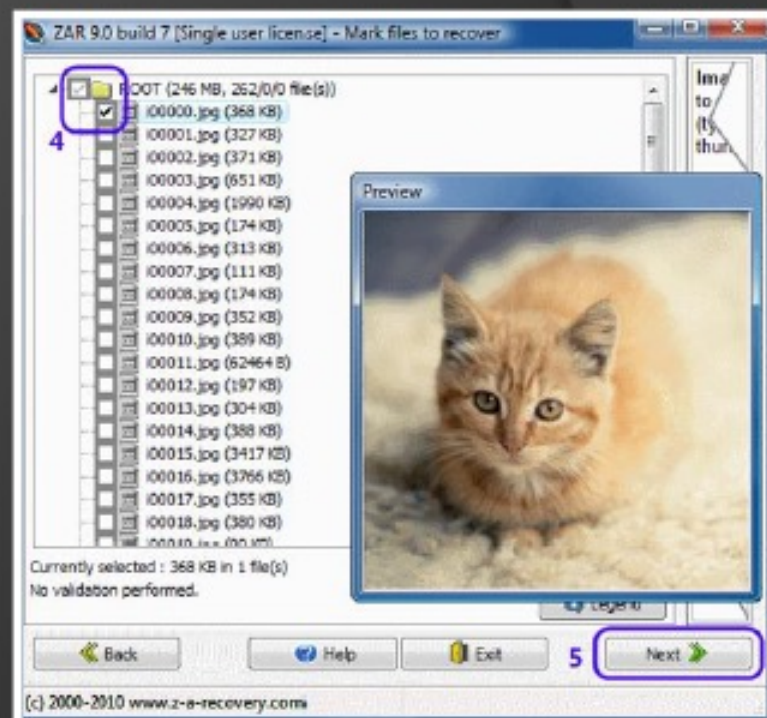
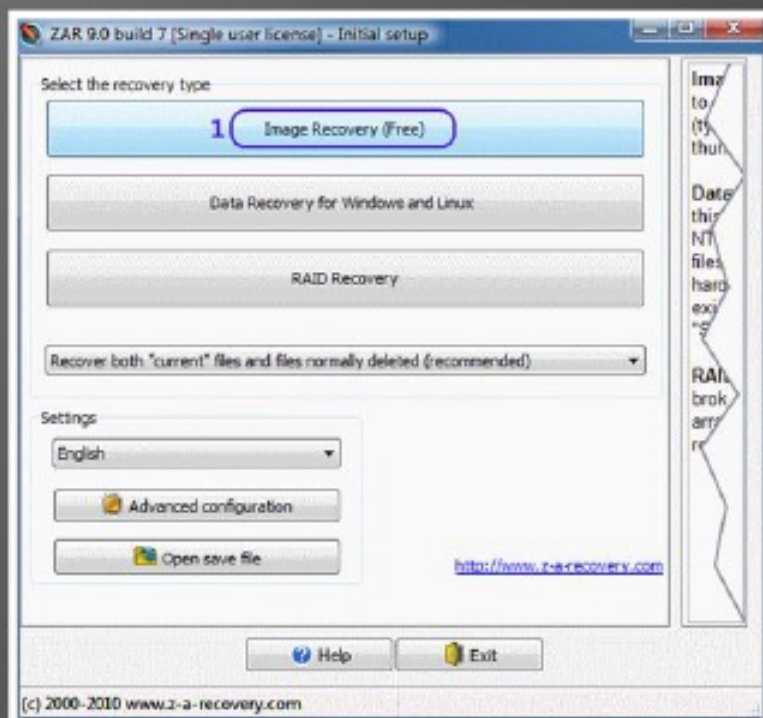
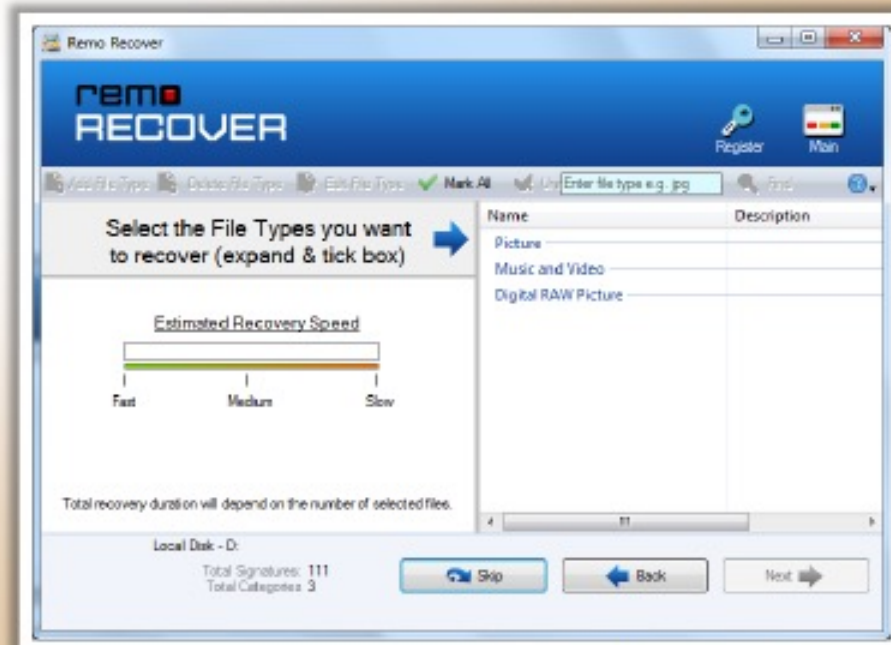
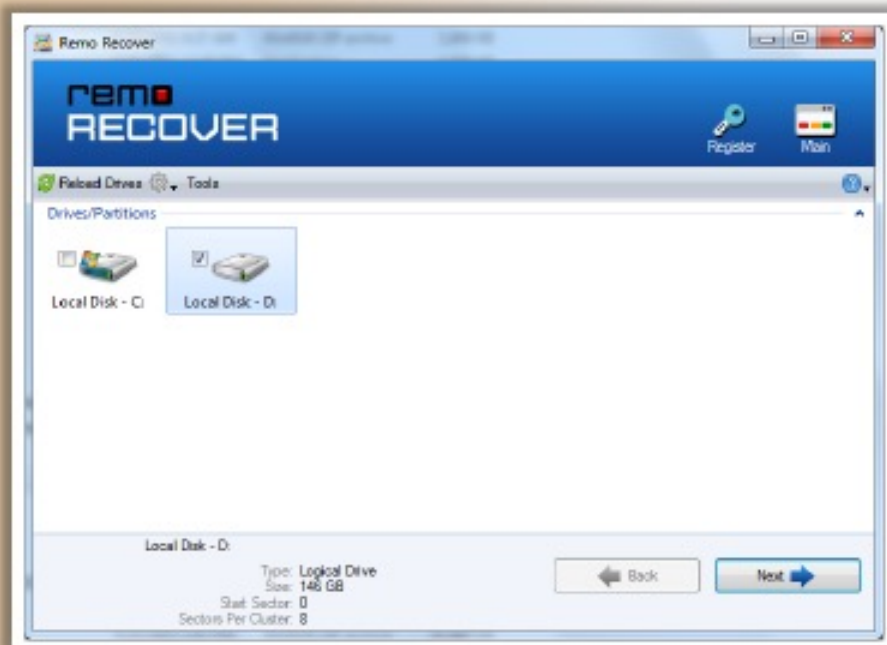


Photo Recovery Software

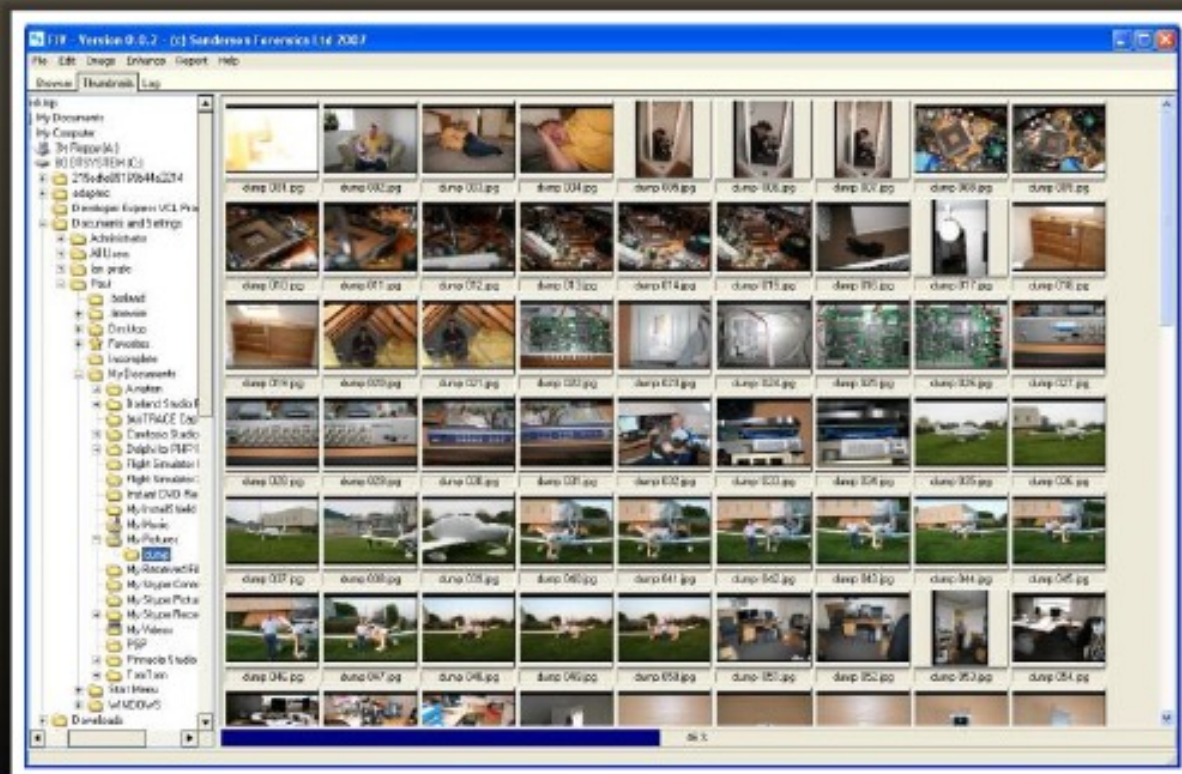
- Photo Recovery Software **recovers digital photos** from USB drives, flash cards, SD cards, CF cards, iPods, hard drives and FireWire drives
- It supports recovery of digital photos from all **Windows accessible partitions**



<http://www.photorecoverysoftware.org>

Forensic Image Viewer

- Forensic Image Viewer (FIV) is an in-development tool for the **processing** and **reporting** of still images (JPG's, PNG's, GIF's etc.)



File Finder

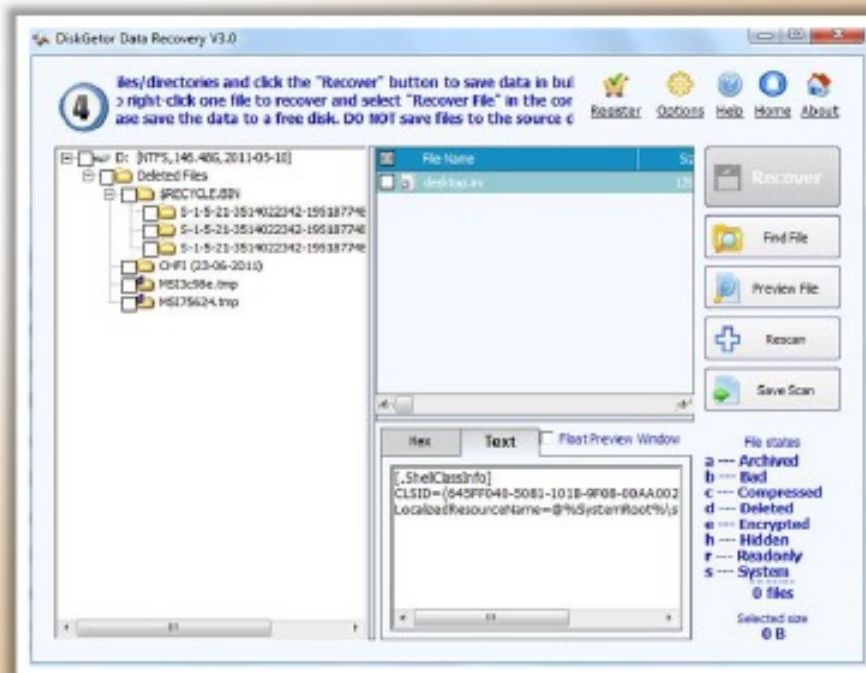
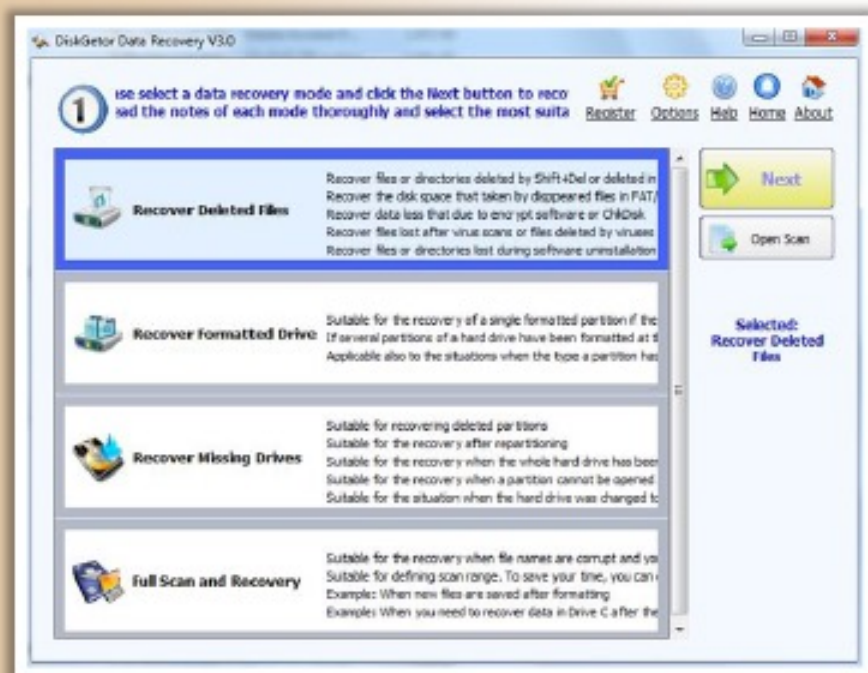
- File Finder **recovers lost or deleted files** on any computer or digital media
- It recovers **image data deleted** or lost on PCs and memory cards
- It recovers **photos and pictures** from MMC cards, SD cards and other photo memory cards



<http://recover-lost-files.com>

DiskGetor Data Recovery

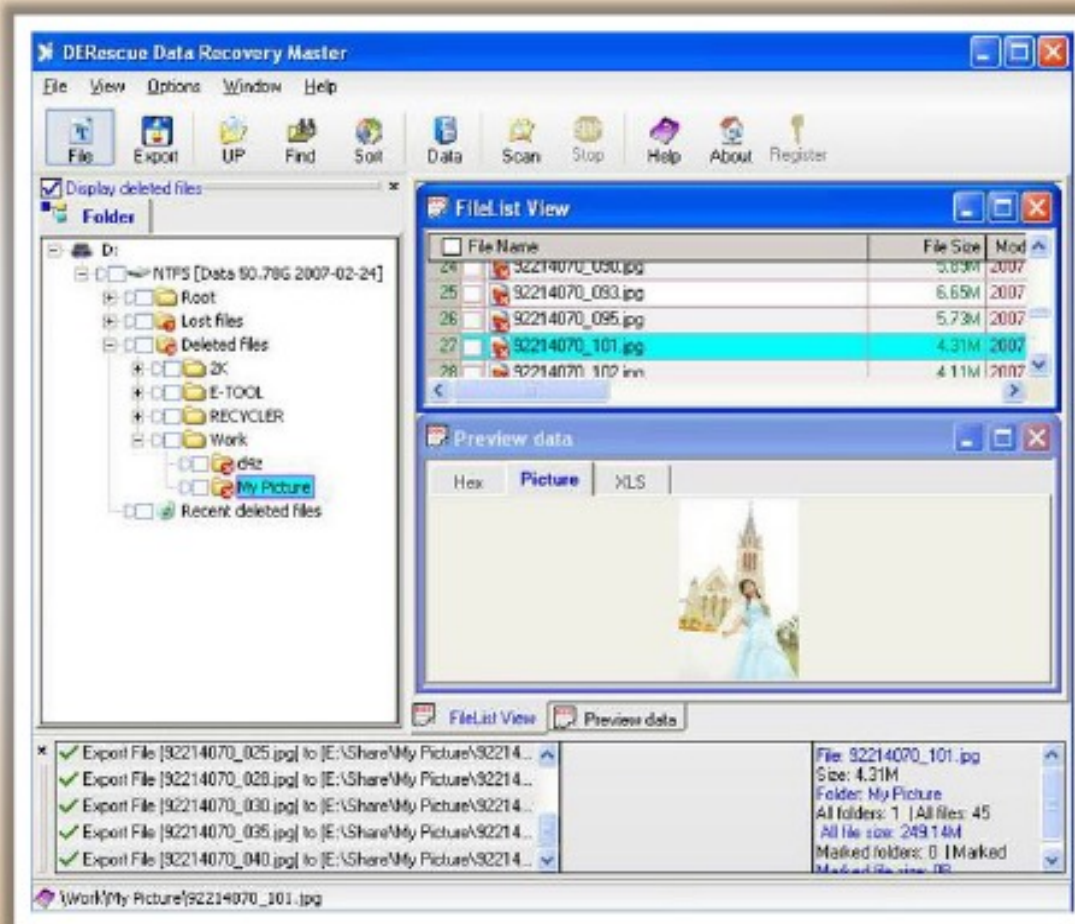
- DiskGetor Data Recovery is **hard drive** data recovery software that recovers formatted data, deleted data, lost data, damaged data from hard drives
- It recovers **image** and **photo** files from the hard drive



<http://www.diskgetor.com>

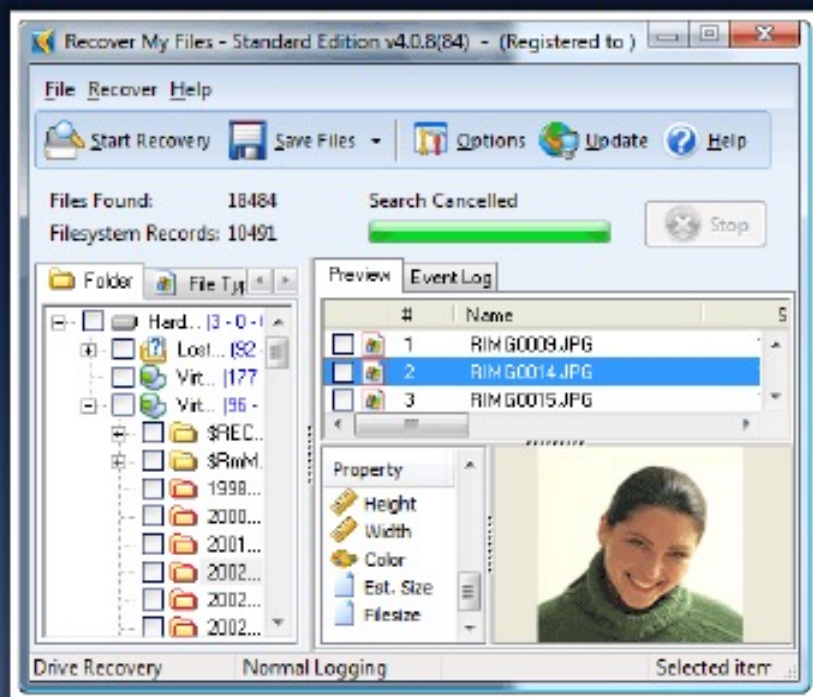
DERescue Data Recovery Master

DERescue Data Recovery Master recovers **deleted**, **formatted** and lost digital photos, images and pictures from any types of **media cards** used by digital cameras or computers



<http://www.derescue.com>

Recover My Files



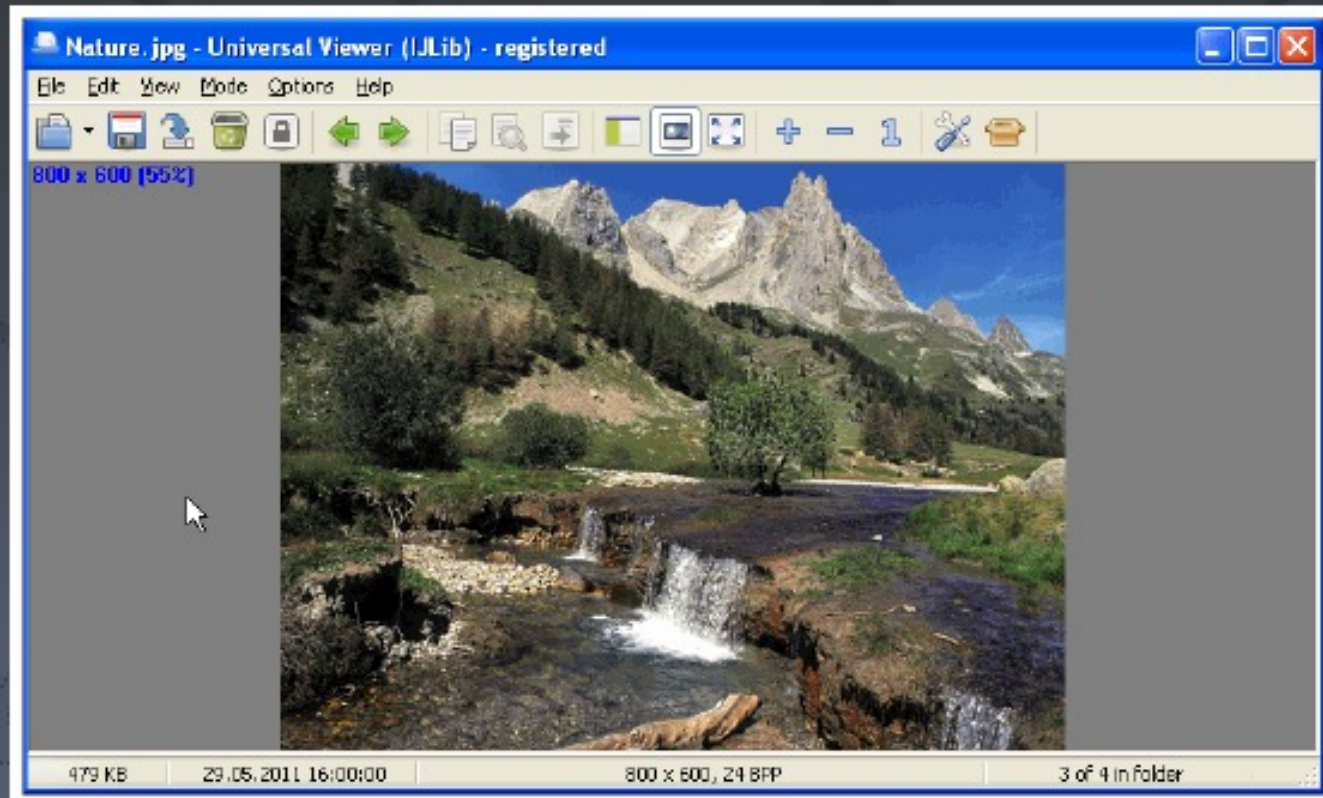
<http://www.recovermyfiles.com>

- Recover My Files data recovery software recovers deleted files emptied from the **Windows Recycle Bin**, or lost due to the format or corruption of a hard drive, virus or Trojan infection, unexpected system **shutdown** or **software failure**
- It recovers **deleted** or **lost picture files** (.jpeg, .jpg, .gif and more) from hard drives, camera cards, USB drives, Zip disks, floppy disks or other removable drives and disks



Universal Viewer

- Universal Viewer is an advanced **file viewer** for a wide range of formats
- Supported file formats are images, multimedia, Word, Excel, PDF, RTF, Internet, text, etc.



<http://www.uvviewsoft.com>

Module Summary



- ☐ Steganography is the method of hiding information by embedding messages within other, seemingly harmless messages
- ☐ Steganalysis is the technology that attempts to defeat steganography—by detecting the hidden information and extracting it or destroying it
- ☐ Steganography is used to secure secret communications
- ☐ An image is an artifact that reproduces the likeness of some subject
- ☐ A file format is a “particular way to encode information for storage in a computer file”
- ☐ The standard image file formats include JPEG, GIF, BMP, TAG, and EPS
- ☐ Data compression means encoding the data to take up less storage space and less bandwidth for transmission
- ☐ Data is compressed by using a complex algorithm to reduce the size of a file
- ☐ Lossy compression compresses data permanently by removing information contained in the file

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www.glasbergen.com



“Is there a file compression program that will help me squeeze 12 hours of work into an 8 hour schedule?”

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**“Someone sent me an e-mail attachment. From the size,
I’m guessing it’s either furniture or some sort of farm animal!”**